MSE 494/5 – AY 2025–26

Batch of 100 Sample Weekly Progress Reports (AI Training Dataset)

Report 1

1. Group Number and Title: Group 1 – Recyclable Polymer Blend Development

2. Date: March 29, 2026

3. HCD Spaces Engaged: Prototype, Ideate

4. Activities Completed

- Activity 1: Conducted work related to Ideate space including data collection, analysis, concept refinement, or material testing.
- Activity 2: Conducted work related to Prototype space including data collection, analysis, concept refinement, or material testing.
- Activity 3: Conducted work related to Ideate space including data collection, analysis, concept refinement, or material testing.
- Activity 4: Conducted work related to Prototype space including data collection, analysis, concept refinement, or material testing.
- Activity 5: Conducted work related to Prototype space including data collection, analysis, concept refinement, or material testing.
- Activity 6: Conducted work related to Prototype space including data collection, analysis, concept refinement, or material testing.

5. Expanded Description

The team worked on assigned tasks and made incremental progress. Limited stakeholder engagement occurred. Decisions were primarily based on technical feasibility.

6. Outstanding Challenges

- Increase stakeholder validation.
- Clarify cost constraints.
- Need additional testing data.

7. Curiosity Development (EM)

The team asked clarification questions and followed provided guidance.

8. Making Connections (EM)

Connections were primarily within course materials.

9. Creating Value (EM)

Progress contributed to completing project milestones.

10. Feedback Addressed

Instructor feedback was incorporated into the next iteration cycle and influenced design decisions.

Report 2

1. Group Number and Title: Group 6 – Smart Bio-Composite Packaging

2. Date: February 28, 2026

3. HCD Spaces Engaged: Implement, Prototype

4. Activities Completed

- Activity 1: Conducted work related to Implement space including data collection, analysis, concept refinement, or material testing.
- Activity 2: Conducted work related to Prototype space including data collection, analysis, concept refinement, or material testing.
- Activity 3: Conducted work related to Prototype space including data collection, analysis, concept refinement, or material testing.

5. Expanded Description

The team deeply integrated stakeholder insights with materials testing. Iterative prototyping informed evidence-based decisions. Trade-offs between performance, manufacturability, and user needs were systematically evaluated.

6. Outstanding Challenges

- Clarify cost constraints.
- Improve prototype durability.
- Need additional testing data.

7. Curiosity Development (EM)

The team demonstrated inquisitiveness, contrarian thinking, experimentation, and comfort with ambiguity while iterating multiple prototype pathways.

8. Making Connections (EM)

The team applied systems thinking, creativity, and strategic risk management to integrate user insights, engineering constraints, and scalability considerations.

9. Creating Value (EM)

The team demonstrated customer-centric thinking, value awareness, persistence, and continuous improvement while refining prototypes to maximize user impact.

10. Feedback Addressed

Instructor feedback was incorporated into the next iteration cycle and influenced design decisions.

Report 3

1. Group Number and Title: Group 6 – Smart Bio-Composite Packaging

2. Date: March 12, 2026

3. HCD Spaces Engaged: Implement, Prototype, Synthesize, Ideate

4. Activities Completed

- Activity 1: Conducted work related to Synthesize space including data collection, analysis, concept refinement, or material testing.
- Activity 2: Conducted work related to Synthesize space including data collection, analysis, concept refinement, or material testing.
- Activity 3: Conducted work related to Ideate space including data collection, analysis, concept refinement, or material testing.
- Activity 4: Conducted work related to Prototype space including data collection, analysis, concept refinement, or material testing.
- Activity 5: Conducted work related to Implement space including data collection, analysis, concept refinement, or material testing.

5. Expanded Description

The team balanced technical testing with user considerations. Some stakeholder feedback informed concept refinement. Iterative testing improved performance metrics.

6. Outstanding Challenges

- Need additional testing data.
- Increase stakeholder validation.
- Clarify cost constraints.
- Address manufacturability risks.

7. Curiosity Development (EM)

The team explored alternative material configurations and questioned initial assumptions about performance trade-offs.

8. Making Connections (EM)

The team linked stakeholder needs with materials properties and considered long-term system implications.

9. Creating Value (EM)

The team focused on improving usability while maintaining technical performance.

10. Feedback Addressed

Instructor feedback was incorporated into the next iteration cycle and influenced design decisions.

Report 4

1. Group Number and Title: Group 7 – Antimicrobial Hospital Surface Coating

2. Date: February 12, 2026

3. HCD Spaces Engaged: Understand, Ideate, Implement

4. Activities Completed

- Activity 1: Conducted work related to Implement space including data collection, analysis, concept refinement, or material testing.
- Activity 2: Conducted work related to Implement space including data collection, analysis, concept refinement, or material testing.
- Activity 3: Conducted work related to Understand space including data collection, analysis, concept refinement, or material testing.
- Activity 4: Conducted work related to Implement space including data collection, analysis, concept refinement, or material testing.
- Activity 5: Conducted work related to Ideate space including data collection, analysis, concept refinement, or material testing.
- Activity 6: Conducted work related to Understand space including data collection, analysis, concept refinement, or material testing.

5. Expanded Description

The team balanced technical testing with user considerations. Some stakeholder feedback informed concept refinement. Iterative testing improved performance metrics.

6. Outstanding Challenges

- Clarify cost constraints.
- Address manufacturability risks.
- Need additional testing data.
- Improve prototype durability.

7. Curiosity Development (EM)

The team explored alternative material configurations and questioned initial assumptions about performance trade-offs.

8. Making Connections (EM)

The team linked stakeholder needs with materials properties and considered long-term system implications.

9. Creating Value (EM)

The team focused on improving usability while maintaining technical performance.

10. Feedback Addressed

Instructor feedback was incorporated into the next iteration cycle and influenced design decisions.

Report 5

1. Group Number and Title: Group 19 – Wearable Thermal Regulation System

2. Date: February 07, 2026

3. HCD Spaces Engaged: Implement, Prototype

4. Activities Completed

- Activity 1: Conducted work related to Implement space including data collection, analysis, concept refinement, or material testing.
- Activity 2: Conducted work related to Prototype space including data collection, analysis, concept refinement, or material testing.
- Activity 3: Conducted work related to Implement space including data collection, analysis, concept refinement, or material testing.
- Activity 4: Conducted work related to Implement space including data collection, analysis, concept refinement, or material testing.

5. Expanded Description

The team deeply integrated stakeholder insights with materials testing. Iterative prototyping informed evidence-based decisions. Trade-offs between performance, manufacturability, and user needs were systematically evaluated.

6. Outstanding Challenges

- Clarify cost constraints.
- Improve prototype durability.
- Need additional testing data.

7. Curiosity Development (EM)

The team demonstrated inquisitiveness, contrarian thinking, experimentation, and comfort with ambiguity while iterating multiple prototype pathways.

8. Making Connections (EM)

The team applied systems thinking, creativity, and strategic risk management to integrate user insights, engineering constraints, and scalability considerations.

9. Creating Value (EM)

The team demonstrated customer-centric thinking, value awareness, persistence, and continuous improvement while refining prototypes to maximize user impact.

10. Feedback Addressed

Instructor feedback was incorporated into the next iteration cycle and influenced design decisions.

Report 6

1. Group Number and Title: Group 10 – Recyclable Polymer Blend Development

2. Date: February 07, 2026

3. HCD Spaces Engaged: Implement, Prototype, Synthesize

4. Activities Completed

- Activity 1: Conducted work related to Implement space including data collection, analysis, concept refinement, or material testing.
- Activity 2: Conducted work related to Implement space including data collection, analysis, concept refinement, or material testing.
- Activity 3: Conducted work related to Prototype space including data collection, analysis, concept refinement, or material testing.
- Activity 4: Conducted work related to Implement space including data collection, analysis, concept refinement, or material testing.
- Activity 5: Conducted work related to Prototype space including data collection, analysis, concept refinement, or material testing.
- Activity 6: Conducted work related to Prototype space including data collection, analysis, concept refinement, or material testing.

5. Expanded Description

The team deeply integrated stakeholder insights with materials testing. Iterative prototyping informed evidence-based decisions. Trade-offs between performance, manufacturability, and user needs were systematically evaluated.

6. Outstanding Challenges

- Increase stakeholder validation.
- Address manufacturability risks.
- Clarify cost constraints.
- Improve prototype durability.

7. Curiosity Development (EM)

The team demonstrated inquisitiveness, contrarian thinking, experimentation, and comfort with ambiguity while iterating multiple prototype pathways.

8. Making Connections (EM)

The team applied systems thinking, creativity, and strategic risk management to integrate user insights, engineering constraints, and scalability considerations.

9. Creating Value (EM)

The team demonstrated customer-centric thinking, value awareness, persistence, and continuous improvement while refining prototypes to maximize user impact.

10. Feedback Addressed

Instructor feedback was incorporated into the next iteration cycle and influenced design decisions.

Report 7

1. Group Number and Title: Group 14 – Smart Bio-Composite Packaging

2. Date: February 16, 2026

3. HCD Spaces Engaged: Implement, Ideate, Understand, Prototype, Synthesize

4. Activities Completed

- Activity 1: Conducted work related to Prototype space including data collection, analysis, concept refinement, or material testing.
- Activity 2: Conducted work related to Synthesize space including data collection, analysis, concept refinement, or material testing.
- Activity 3: Conducted work related to Understand space including data collection, analysis, concept refinement, or material testing.
- Activity 4: Conducted work related to Synthesize space including data collection, analysis, concept refinement, or material testing.

5. Expanded Description

The team balanced technical testing with user considerations. Some stakeholder feedback informed concept refinement. Iterative testing improved performance metrics.

6. Outstanding Challenges

- Clarify cost constraints.
- Need additional testing data.
- Improve prototype durability.
- Address manufacturability risks.

7. Curiosity Development (EM)

The team explored alternative material configurations and questioned initial assumptions about performance trade-offs.

8. Making Connections (EM)

The team linked stakeholder needs with materials properties and considered long-term system implications.

9. Creating Value (EM)

The team focused on improving usability while maintaining technical performance.

10. Feedback Addressed

Instructor feedback was incorporated into the next iteration cycle and influenced design decisions.

Report 8

1. Group Number and Title: Group 12 – Sustainable Concrete Additive

2. Date: February 11, 2026

3. HCD Spaces Engaged: Synthesize, Implement

4. Activities Completed

- Activity 1: Conducted work related to Implement space including data collection, analysis, concept refinement, or material testing.
- Activity 2: Conducted work related to Synthesize space including data collection, analysis, concept refinement, or material testing.
- Activity 3: Conducted work related to Synthesize space including data collection, analysis, concept refinement, or material testing.
- Activity 4: Conducted work related to Implement space including data collection, analysis, concept refinement, or material testing.
- Activity 5: Conducted work related to Synthesize space including data collection, analysis, concept refinement, or material testing.
- Activity 6: Conducted work related to Implement space including data collection, analysis, concept refinement, or material testing.

5. Expanded Description

The team deeply integrated stakeholder insights with materials testing. Iterative prototyping informed evidence-based decisions. Trade-offs between performance, manufacturability, and user needs were systematically evaluated.

6. Outstanding Challenges

- Address manufacturability risks.
- Clarify cost constraints.
- Improve prototype durability.

7. Curiosity Development (EM)

The team demonstrated inquisitiveness, contrarian thinking, experimentation, and comfort with ambiguity while iterating multiple prototype pathways.

8. Making Connections (EM)

The team applied systems thinking, creativity, and strategic risk management to integrate user insights, engineering constraints, and scalability considerations.

9. Creating Value (EM)

The team demonstrated customer-centric thinking, value awareness, persistence, and continuous improvement while refining prototypes to maximize user impact.

10. Feedback Addressed

Instructor feedback was incorporated into the next iteration cycle and influenced design decisions.

Report 9

1. Group Number and Title: Group 5 – Phase-Change Cooling Vest

2. Date: February 26, 2026

3. HCD Spaces Engaged: Understand, Synthesize

4. Activities Completed

- Activity 1: Conducted work related to Synthesize space including data collection, analysis, concept refinement, or material testing.
- Activity 2: Conducted work related to Understand space including data collection, analysis, concept refinement, or material testing.
- Activity 3: Conducted work related to Synthesize space including data collection, analysis, concept refinement, or material testing.
- Activity 4: Conducted work related to Understand space including data collection, analysis, concept refinement, or material testing.
- Activity 5: Conducted work related to Understand space including data collection, analysis, concept refinement, or material testing.
- Activity 6: Conducted work related to Understand space including data collection, analysis, concept refinement, or material testing.

5. Expanded Description

The team deeply integrated stakeholder insights with materials testing. Iterative prototyping informed evidence-based decisions. Trade-offs between performance, manufacturability, and user needs were systematically evaluated.

6. Outstanding Challenges

- Need additional testing data.
- Increase stakeholder validation.
- Improve prototype durability.

7. Curiosity Development (EM)

The team demonstrated inquisitiveness, contrarian thinking, experimentation, and comfort with ambiguity while iterating multiple prototype pathways.

8. Making Connections (EM)

The team applied systems thinking, creativity, and strategic risk management to integrate user insights, engineering constraints, and scalability considerations.

9. Creating Value (EM)

The team demonstrated customer-centric thinking, value awareness, persistence, and continuous improvement while refining prototypes to maximize user impact.

10. Feedback Addressed

Instructor feedback was incorporated into the next iteration cycle and influenced design decisions.

Report 10

1. Group Number and Title: Group 11 – Lightweight Aerospace Alloy Redesign

2. Date: February 11, 2026

3. HCD Spaces Engaged: Implement, Synthesize

4. Activities Completed

- Activity 1: Conducted work related to Implement space including data collection, analysis, concept refinement, or material testing.
- Activity 2: Conducted work related to Synthesize space including data collection, analysis, concept refinement, or material testing.
- Activity 3: Conducted work related to Implement space including data collection, analysis, concept refinement, or material testing.
- Activity 4: Conducted work related to Synthesize space including data collection, analysis, concept refinement, or material testing.
- Activity 5: Conducted work related to Synthesize space including data collection, analysis, concept refinement, or material testing.
- Activity 6: Conducted work related to Implement space including data collection, analysis, concept refinement, or material testing.

5. Expanded Description

The team deeply integrated stakeholder insights with materials testing. Iterative prototyping informed evidence-based decisions. Trade-offs between performance, manufacturability, and user needs were systematically evaluated.

6. Outstanding Challenges

- Need additional testing data.
- Address manufacturability risks.
- Increase stakeholder validation.
- Improve prototype durability.

7. Curiosity Development (EM)

The team demonstrated inquisitiveness, contrarian thinking, experimentation, and comfort with ambiguity while iterating multiple prototype pathways.

8. Making Connections (EM)

The team applied systems thinking, creativity, and strategic risk management to integrate user insights, engineering constraints, and scalability considerations.

9. Creating Value (EM)

The team demonstrated customer-centric thinking, value awareness, persistence, and continuous improvement while refining prototypes to maximize user impact.

10. Feedback Addressed

Instructor feedback was incorporated into the next iteration cycle and influenced design decisions.

Report 11

1. Group Number and Title: Group 14 – Low-Cost Water Filtration Membrane

2. Date: March 21, 2026

3. HCD Spaces Engaged: Implement, Synthesize, Understand, Ideate, Prototype

4. Activities Completed

- Activity 1: Conducted work related to Ideate space including data collection, analysis, concept refinement, or material testing.
- Activity 2: Conducted work related to Ideate space including data collection, analysis, concept refinement, or material testing.
- Activity 3: Conducted work related to Synthesize space including data collection, analysis, concept refinement, or material testing.

5. Expanded Description

The team balanced technical testing with user considerations. Some stakeholder feedback informed concept refinement. Iterative testing improved performance metrics.

6. Outstanding Challenges

- Address manufacturability risks.
- Need additional testing data.
- Improve prototype durability.
- Increase stakeholder validation.

7. Curiosity Development (EM)

The team explored alternative material configurations and questioned initial assumptions about performance trade-offs.

8. Making Connections (EM)

The team linked stakeholder needs with materials properties and considered long-term system implications.

9. Creating Value (EM)

The team focused on improving usability while maintaining technical performance.

10. Feedback Addressed

Instructor feedback was incorporated into the next iteration cycle and influenced design decisions.

Report 12

1. Group Number and Title: Group 12 – Recyclable Polymer Blend Development

2. Date: February 02, 2026

3. HCD Spaces Engaged: Synthesize, Understand, Prototype

4. Activities Completed

- Activity 1: Conducted work related to Synthesize space including data collection, analysis, concept refinement, or material testing.
- Activity 2: Conducted work related to Prototype space including data collection, analysis, concept refinement, or material testing.
- Activity 3: Conducted work related to Prototype space including data collection, analysis, concept refinement, or material testing.

5. Expanded Description

The team worked on assigned tasks and made incremental progress. Limited stakeholder engagement occurred. Decisions were primarily based on technical feasibility.

6. Outstanding Challenges

- Need additional testing data.
- Clarify cost constraints.
- Improve prototype durability.

7. Curiosity Development (EM)

The team asked clarification questions and followed provided guidance.

8. Making Connections (EM)

Connections were primarily within course materials.

9. Creating Value (EM)

Progress contributed to completing project milestones.

10. Feedback Addressed

Instructor feedback was incorporated into the next iteration cycle and influenced design decisions.

Report 13

1. Group Number and Title: Group 12 – Recyclable Polymer Blend Development

2. Date: March 25, 2026

3. HCD Spaces Engaged: Ideate, Synthesize, Understand, Prototype, Implement

4. Activities Completed

- Activity 1: Conducted work related to Implement space including data collection, analysis, concept refinement, or material testing.
- Activity 2: Conducted work related to Synthesize space including data collection, analysis, concept refinement, or material testing.
- Activity 3: Conducted work related to Understand space including data collection, analysis, concept refinement, or material testing.
- Activity 4: Conducted work related to Prototype space including data collection, analysis, concept refinement, or material testing.

5. Expanded Description

The team balanced technical testing with user considerations. Some stakeholder feedback informed concept refinement. Iterative testing improved performance metrics.

6. Outstanding Challenges

- Improve prototype durability.
- Clarify cost constraints.
- Address manufacturability risks.

7. Curiosity Development (EM)

The team explored alternative material configurations and questioned initial assumptions about performance trade-offs.

8. Making Connections (EM)

The team linked stakeholder needs with materials properties and considered long-term system implications.

9. Creating Value (EM)

The team focused on improving usability while maintaining technical performance.

10. Feedback Addressed

Instructor feedback was incorporated into the next iteration cycle and influenced design decisions.

Report 14

1. Group Number and Title: Group 15 – Phase-Change Cooling Vest

2. Date: February 13, 2026

3. HCD Spaces Engaged: Synthesize, Implement

4. Activities Completed

- Activity 1: Conducted work related to Implement space including data collection, analysis, concept refinement, or material testing.
- Activity 2: Conducted work related to Implement space including data collection, analysis, concept refinement, or material testing.
- Activity 3: Conducted work related to Synthesize space including data collection, analysis, concept refinement, or material testing.
- Activity 4: Conducted work related to Implement space including data collection, analysis, concept refinement, or material testing.
- Activity 5: Conducted work related to Synthesize space including data collection, analysis, concept refinement, or material testing.

5. Expanded Description

The team worked on assigned tasks and made incremental progress. Limited stakeholder engagement occurred. Decisions were primarily based on technical feasibility.

6. Outstanding Challenges

- Improve prototype durability.
- Address manufacturability risks.

7. Curiosity Development (EM)

The team asked clarification questions and followed provided guidance.

8. Making Connections (EM)

Connections were primarily within course materials.

9. Creating Value (EM)

Progress contributed to completing project milestones.

10. Feedback Addressed

Instructor feedback was incorporated into the next iteration cycle and influenced design decisions.

Report 15

1. Group Number and Title: Group 3 – Low-Cost Water Filtration Membrane

2. Date: March 04, 2026

3. HCD Spaces Engaged: Synthesize, Implement, Prototype, Ideate, Understand

4. Activities Completed

- Activity 1: Conducted work related to Ideate space including data collection, analysis, concept refinement, or material testing.
- Activity 2: Conducted work related to Understand space including data collection, analysis, concept refinement, or material testing.
- Activity 3: Conducted work related to Prototype space including data collection, analysis, concept refinement, or material testing.
- Activity 4: Conducted work related to Ideate space including data collection, analysis, concept refinement, or material testing.
- Activity 5: Conducted work related to Understand space including data collection, analysis, concept refinement, or material testing.
- Activity 6: Conducted work related to Prototype space including data collection, analysis, concept refinement, or material testing.

5. Expanded Description

The team worked on assigned tasks and made incremental progress. Limited stakeholder engagement occurred. Decisions were primarily based on technical feasibility.

6. Outstanding Challenges

- Address manufacturability risks.
- Clarify cost constraints.
- Increase stakeholder validation.
- Need additional testing data.

7. Curiosity Development (EM)

The team asked clarification questions and followed provided guidance.

8. Making Connections (EM)

Connections were primarily within course materials.

9. Creating Value (EM)

Progress contributed to completing project milestones.

10. Feedback Addressed

Instructor feedback was incorporated into the next iteration cycle and influenced design decisions.

Report 16

1. Group Number and Title: Group 8 – Low-Cost Water Filtration Membrane

2. Date: February 28, 2026

3. HCD Spaces Engaged: Understand, Prototype

4. Activities Completed

- Activity 1: Conducted work related to Prototype space including data collection, analysis, concept refinement, or material testing.
- Activity 2: Conducted work related to Understand space including data collection, analysis, concept refinement, or material testing.
- Activity 3: Conducted work related to Understand space including data collection, analysis, concept refinement, or material testing.
- Activity 4: Conducted work related to Understand space including data collection, analysis, concept refinement, or material testing.
- Activity 5: Conducted work related to Understand space including data collection, analysis, concept refinement, or material testing.
- Activity 6: Conducted work related to Prototype space including data collection, analysis, concept refinement, or material testing.

5. Expanded Description

The team worked on assigned tasks and made incremental progress. Limited stakeholder engagement occurred. Decisions were primarily based on technical feasibility.

6. Outstanding Challenges

- Increase stakeholder validation.
- Improve prototype durability.
- Need additional testing data.

7. Curiosity Development (EM)

The team asked clarification questions and followed provided guidance.

8. Making Connections (EM)

Connections were primarily within course materials.

9. Creating Value (EM)

Progress contributed to completing project milestones.

10. Feedback Addressed

Instructor feedback was incorporated into the next iteration cycle and influenced design decisions.

Report 17

1. Group Number and Title: Group 18 – Heat-Resistant Industrial Glove

2. Date: March 02, 2026

3. HCD Spaces Engaged: Understand, Synthesize, Implement

4. Activities Completed

- Activity 1: Conducted work related to Understand space including data collection, analysis, concept refinement, or material testing.
- Activity 2: Conducted work related to Understand space including data collection, analysis, concept refinement, or material testing.
- Activity 3: Conducted work related to Implement space including data collection, analysis, concept refinement, or material testing.
- Activity 4: Conducted work related to Synthesize space including data collection, analysis, concept refinement, or material testing.
- Activity 5: Conducted work related to Implement space including data collection, analysis, concept refinement, or material testing.

5. Expanded Description

The team deeply integrated stakeholder insights with materials testing. Iterative prototyping informed evidence-based decisions. Trade-offs between performance, manufacturability, and user needs were systematically evaluated.

6. Outstanding Challenges

- Address manufacturability risks.
- Increase stakeholder validation.

7. Curiosity Development (EM)

The team demonstrated inquisitiveness, contrarian thinking, experimentation, and comfort with ambiguity while iterating multiple prototype pathways.

8. Making Connections (EM)

The team applied systems thinking, creativity, and strategic risk management to integrate user insights, engineering constraints, and scalability considerations.

9. Creating Value (EM)

The team demonstrated customer-centric thinking, value awareness, persistence, and continuous improvement while refining prototypes to maximize user impact.

10. Feedback Addressed

Instructor feedback was incorporated into the next iteration cycle and influenced design decisions.

Report 18

1. Group Number and Title: Group 18 – Flexible Solar Encapsulation Layer

2. Date: March 09, 2026

3. HCD Spaces Engaged: Ideate, Implement, Prototype, Understand

4. Activities Completed

- Activity 1: Conducted work related to Prototype space including data collection, analysis, concept refinement, or material testing.
- Activity 2: Conducted work related to Prototype space including data collection, analysis, concept refinement, or material testing.
- Activity 3: Conducted work related to Implement space including data collection, analysis, concept refinement, or material testing.
- Activity 4: Conducted work related to Ideate space including data collection, analysis, concept refinement, or material testing.
- Activity 5: Conducted work related to Ideate space including data collection, analysis, concept refinement, or material testing.

5. Expanded Description

The team worked on assigned tasks and made incremental progress. Limited stakeholder engagement occurred. Decisions were primarily based on technical feasibility.

6. Outstanding Challenges

- Need additional testing data.
- Address manufacturability risks.
- Clarify cost constraints.

7. Curiosity Development (EM)

The team asked clarification questions and followed provided guidance.

8. Making Connections (EM)

Connections were primarily within course materials.

9. Creating Value (EM)

Progress contributed to completing project milestones.

10. Feedback Addressed

Instructor feedback was incorporated into the next iteration cycle and influenced design decisions.

Report 19

1. Group Number and Title: Group 18 – Sustainable Concrete Additive

2. Date: February 03, 2026

3. HCD Spaces Engaged: Ideate, Prototype, Synthesize, Understand, Implement

4. Activities Completed

- Activity 1: Conducted work related to Synthesize space including data collection, analysis, concept refinement, or material testing.
- Activity 2: Conducted work related to Prototype space including data collection, analysis, concept refinement, or material testing.
- Activity 3: Conducted work related to Ideate space including data collection, analysis, concept refinement, or material testing.

5. Expanded Description

The team worked on assigned tasks and made incremental progress. Limited stakeholder engagement occurred. Decisions were primarily based on technical feasibility.

6. Outstanding Challenges

- Address manufacturability risks.
- Improve prototype durability.
- Clarify cost constraints.

7. Curiosity Development (EM)

The team asked clarification questions and followed provided guidance.

8. Making Connections (EM)

Connections were primarily within course materials.

9. Creating Value (EM)

Progress contributed to completing project milestones.

10. Feedback Addressed

Instructor feedback was incorporated into the next iteration cycle and influenced design decisions.

Report 20

1. Group Number and Title: Group 25 – Sustainable Concrete Additive

2. Date: March 05, 2026

3. HCD Spaces Engaged: Implement, Ideate

4. Activities Completed

- Activity 1: Conducted work related to Ideate space including data collection, analysis, concept refinement, or material testing.
- Activity 2: Conducted work related to Implement space including data collection, analysis, concept refinement, or material testing.
- Activity 3: Conducted work related to Implement space including data collection, analysis, concept refinement, or material testing.
- Activity 4: Conducted work related to Implement space including data collection, analysis, concept refinement, or material testing.
- Activity 5: Conducted work related to Ideate space including data collection, analysis, concept refinement, or material testing.

5. Expanded Description

The team deeply integrated stakeholder insights with materials testing. Iterative prototyping informed evidence-based decisions. Trade-offs between performance, manufacturability, and user needs were systematically evaluated.

6. Outstanding Challenges

- Increase stakeholder validation.
- Need additional testing data.
- Improve prototype durability.
- Address manufacturability risks.

7. Curiosity Development (EM)

The team demonstrated inquisitiveness, contrarian thinking, experimentation, and comfort with ambiguity while iterating multiple prototype pathways.

8. Making Connections (EM)

The team applied systems thinking, creativity, and strategic risk management to integrate user insights, engineering constraints, and scalability considerations.

9. Creating Value (EM)

The team demonstrated customer-centric thinking, value awareness, persistence, and continuous improvement while refining prototypes to maximize user impact.

10. Feedback Addressed

Instructor feedback was incorporated into the next iteration cycle and influenced design decisions.

Report 21

1. Group Number and Title: Group 22 – Low-Cost Water Filtration Membrane

2. Date: March 16, 2026

3. HCD Spaces Engaged: Ideate, Synthesize, Implement

4. Activities Completed

- Activity 1: Conducted work related to Synthesize space including data collection, analysis, concept refinement, or material testing.
- Activity 2: Conducted work related to Implement space including data collection, analysis, concept refinement, or material testing.
- Activity 3: Conducted work related to Ideate space including data collection, analysis, concept refinement, or material testing.
- Activity 4: Conducted work related to Ideate space including data collection, analysis, concept refinement, or material testing.
- Activity 5: Conducted work related to Implement space including data collection, analysis, concept refinement, or material testing.

5. Expanded Description

The team worked on assigned tasks and made incremental progress. Limited stakeholder engagement occurred. Decisions were primarily based on technical feasibility.

6. Outstanding Challenges

- Increase stakeholder validation.
- Address manufacturability risks.

7. Curiosity Development (EM)

The team asked clarification questions and followed provided guidance.

8. Making Connections (EM)

Connections were primarily within course materials.

9. Creating Value (EM)

Progress contributed to completing project milestones.

10. Feedback Addressed

Instructor feedback was incorporated into the next iteration cycle and influenced design decisions.

Report 22

1. Group Number and Title: Group 13 – Lightweight Aerospace Alloy Redesign

2. Date: March 09, 2026

3. HCD Spaces Engaged: Prototype, Understand, Synthesize

4. Activities Completed

- Activity 1: Conducted work related to Understand space including data collection, analysis, concept refinement, or material testing.
- Activity 2: Conducted work related to Understand space including data collection, analysis, concept refinement, or material testing.
- Activity 3: Conducted work related to Understand space including data collection, analysis, concept refinement, or material testing.
- Activity 4: Conducted work related to Synthesize space including data collection, analysis, concept refinement, or material testing.

5. Expanded Description

The team balanced technical testing with user considerations. Some stakeholder feedback informed concept refinement. Iterative testing improved performance metrics.

6. Outstanding Challenges

- Address manufacturability risks.
- Improve prototype durability.

7. Curiosity Development (EM)

The team explored alternative material configurations and questioned initial assumptions about performance trade-offs.

8. Making Connections (EM)

The team linked stakeholder needs with materials properties and considered long-term system implications.

9. Creating Value (EM)

The team focused on improving usability while maintaining technical performance.

10. Feedback Addressed

Instructor feedback was incorporated into the next iteration cycle and influenced design decisions.

Report 23

1. Group Number and Title: Group 3 – Lightweight Aerospace Alloy Redesign

2. Date: March 27, 2026

3. HCD Spaces Engaged: Implement, Synthesize, Ideate, Prototype

4. Activities Completed

- Activity 1: Conducted work related to Implement space including data collection, analysis, concept refinement, or material testing.
- Activity 2: Conducted work related to Implement space including data collection, analysis, concept refinement, or material testing.
- Activity 3: Conducted work related to Implement space including data collection, analysis, concept refinement, or material testing.
- Activity 4: Conducted work related to Ideate space including data collection, analysis, concept refinement, or material testing.

5. Expanded Description

The team deeply integrated stakeholder insights with materials testing. Iterative prototyping informed evidence-based decisions. Trade-offs between performance, manufacturability, and user needs were systematically evaluated.

6. Outstanding Challenges

- Address manufacturability risks.
- Improve prototype durability.
- Need additional testing data.

7. Curiosity Development (EM)

The team demonstrated inquisitiveness, contrarian thinking, experimentation, and comfort with ambiguity while iterating multiple prototype pathways.

8. Making Connections (EM)

The team applied systems thinking, creativity, and strategic risk management to integrate user insights, engineering constraints, and scalability considerations.

9. Creating Value (EM)

The team demonstrated customer-centric thinking, value awareness, persistence, and continuous improvement while refining prototypes to maximize user impact.

10. Feedback Addressed

Instructor feedback was incorporated into the next iteration cycle and influenced design decisions.

Report 24

1. Group Number and Title: Group 25 – Sustainable Concrete Additive

2. Date: February 21, 2026

3. HCD Spaces Engaged: Synthesize, Prototype, Implement, Ideate, Understand

4. Activities Completed

- Activity 1: Conducted work related to Implement space including data collection, analysis, concept refinement, or material testing.
- Activity 2: Conducted work related to Prototype space including data collection, analysis, concept refinement, or material testing.
- Activity 3: Conducted work related to Prototype space including data collection, analysis, concept refinement, or material testing.
- Activity 4: Conducted work related to Prototype space including data collection, analysis, concept refinement, or material testing.

5. Expanded Description

The team deeply integrated stakeholder insights with materials testing. Iterative prototyping informed evidence-based decisions. Trade-offs between performance, manufacturability, and user needs were systematically evaluated.

6. Outstanding Challenges

- Need additional testing data.
- Improve prototype durability.

7. Curiosity Development (EM)

The team demonstrated inquisitiveness, contrarian thinking, experimentation, and comfort with ambiguity while iterating multiple prototype pathways.

8. Making Connections (EM)

The team applied systems thinking, creativity, and strategic risk management to integrate user insights, engineering constraints, and scalability considerations.

9. Creating Value (EM)

The team demonstrated customer-centric thinking, value awareness, persistence, and continuous improvement while refining prototypes to maximize user impact.

10. Feedback Addressed

Instructor feedback was incorporated into the next iteration cycle and influenced design decisions.

Report 25

1. Group Number and Title: Group 13 – Antimicrobial Hospital Surface Coating

2. Date: March 21, 2026

3. HCD Spaces Engaged: Understand, Synthesize

4. Activities Completed

- Activity 1: Conducted work related to Understand space including data collection, analysis, concept refinement, or material testing.
- Activity 2: Conducted work related to Understand space including data collection, analysis, concept refinement, or material testing.
- Activity 3: Conducted work related to Understand space including data collection, analysis, concept refinement, or material testing.
- Activity 4: Conducted work related to Synthesize space including data collection, analysis, concept refinement, or material testing.

5. Expanded Description

The team worked on assigned tasks and made incremental progress. Limited stakeholder engagement occurred. Decisions were primarily based on technical feasibility.

6. Outstanding Challenges

- Clarify cost constraints.
- Need additional testing data.
- Address manufacturability risks.

7. Curiosity Development (EM)

The team asked clarification questions and followed provided guidance.

8. Making Connections (EM)

Connections were primarily within course materials.

9. Creating Value (EM)

Progress contributed to completing project milestones.

10. Feedback Addressed

Instructor feedback was incorporated into the next iteration cycle and influenced design decisions.

Report 26

1. Group Number and Title: Group 24 – Smart Bio-Composite Packaging

2. Date: March 24, 2026

3. HCD Spaces Engaged: Understand, Ideate, Implement

4. Activities Completed

- Activity 1: Conducted work related to Ideate space including data collection, analysis, concept refinement, or material testing.
- Activity 2: Conducted work related to Understand space including data collection, analysis, concept refinement, or material testing.
- Activity 3: Conducted work related to Understand space including data collection, analysis, concept refinement, or material testing.
- Activity 4: Conducted work related to Understand space including data collection, analysis, concept refinement, or material testing.
- Activity 5: Conducted work related to Understand space including data collection, analysis, concept refinement, or material testing.

5. Expanded Description

The team deeply integrated stakeholder insights with materials testing. Iterative prototyping informed evidence-based decisions. Trade-offs between performance, manufacturability, and user needs were systematically evaluated.

6. Outstanding Challenges

- Increase stakeholder validation.
- Need additional testing data.
- Improve prototype durability.
- Address manufacturability risks.

7. Curiosity Development (EM)

The team demonstrated inquisitiveness, contrarian thinking, experimentation, and comfort with ambiguity while iterating multiple prototype pathways.

8. Making Connections (EM)

The team applied systems thinking, creativity, and strategic risk management to integrate user insights, engineering constraints, and scalability considerations.

9. Creating Value (EM)

The team demonstrated customer-centric thinking, value awareness, persistence, and continuous improvement while refining prototypes to maximize user impact.

10. Feedback Addressed

Instructor feedback was incorporated into the next iteration cycle and influenced design decisions.

Report 27

1. Group Number and Title: Group 8 – Recyclable Polymer Blend Development

2. Date: February 10, 2026

3. HCD Spaces Engaged: Understand, Prototype, Implement, Synthesize

4. Activities Completed

- Activity 1: Conducted work related to Understand space including data collection, analysis, concept refinement, or material testing.
- Activity 2: Conducted work related to Understand space including data collection, analysis, concept refinement, or material testing.
- Activity 3: Conducted work related to Implement space including data collection, analysis, concept refinement, or material testing.
- Activity 4: Conducted work related to Understand space including data collection, analysis, concept refinement, or material testing.
- Activity 5: Conducted work related to Implement space including data collection, analysis, concept refinement, or material testing.

5. Expanded Description

The team balanced technical testing with user considerations. Some stakeholder feedback informed concept refinement. Iterative testing improved performance metrics.

6. Outstanding Challenges

- Increase stakeholder validation.
- Need additional testing data.
- Improve prototype durability.
- Clarify cost constraints.

7. Curiosity Development (EM)

The team explored alternative material configurations and questioned initial assumptions about performance trade-offs.

8. Making Connections (EM)

The team linked stakeholder needs with materials properties and considered long-term system implications.

9. Creating Value (EM)

The team focused on improving usability while maintaining technical performance.

10. Feedback Addressed

Instructor feedback was incorporated into the next iteration cycle and influenced design decisions.

Report 28

1. Group Number and Title: Group 3 – Antimicrobial Hospital Surface Coating

2. Date: March 16, 2026

3. HCD Spaces Engaged: Implement, Prototype, Synthesize, Ideate, Understand

4. Activities Completed

- Activity 1: Conducted work related to Synthesize space including data collection, analysis, concept refinement, or material testing.
- Activity 2: Conducted work related to Implement space including data collection, analysis, concept refinement, or material testing.
- Activity 3: Conducted work related to Understand space including data collection, analysis, concept refinement, or material testing.
- Activity 4: Conducted work related to Understand space including data collection, analysis, concept refinement, or material testing.
- Activity 5: Conducted work related to Prototype space including data collection, analysis, concept refinement, or material testing.
- Activity 6: Conducted work related to Prototype space including data collection, analysis, concept refinement, or material testing.

5. Expanded Description

The team deeply integrated stakeholder insights with materials testing. Iterative prototyping informed evidence-based decisions. Trade-offs between performance, manufacturability, and user needs were systematically evaluated.

6. Outstanding Challenges

- Need additional testing data.
- Address manufacturability risks.

7. Curiosity Development (EM)

The team demonstrated inquisitiveness, contrarian thinking, experimentation, and comfort with ambiguity while iterating multiple prototype pathways.

8. Making Connections (EM)

The team applied systems thinking, creativity, and strategic risk management to integrate user insights, engineering constraints, and scalability considerations.

9. Creating Value (EM)

The team demonstrated customer-centric thinking, value awareness, persistence, and continuous improvement while refining prototypes to maximize user impact.

10. Feedback Addressed

Instructor feedback was incorporated into the next iteration cycle and influenced design decisions.

Report 29

1. Group Number and Title: Group 23 – Antimicrobial Hospital Surface Coating

2. Date: March 17, 2026

3. HCD Spaces Engaged: Ideate, Understand, Prototype, Implement, Synthesize

4. Activities Completed

- Activity 1: Conducted work related to Prototype space including data collection, analysis, concept refinement, or material testing.
- Activity 2: Conducted work related to Synthesize space including data collection, analysis, concept refinement, or material testing.
- Activity 3: Conducted work related to Understand space including data collection, analysis, concept refinement, or material testing.
- Activity 4: Conducted work related to Ideate space including data collection, analysis, concept refinement, or material testing.
- Activity 5: Conducted work related to Synthesize space including data collection, analysis, concept refinement, or material testing.

5. Expanded Description

The team balanced technical testing with user considerations. Some stakeholder feedback informed concept refinement. Iterative testing improved performance metrics.

6. Outstanding Challenges

- Improve prototype durability.
- Need additional testing data.
- Clarify cost constraints.
- Increase stakeholder validation.

7. Curiosity Development (EM)

The team explored alternative material configurations and questioned initial assumptions about performance trade-offs.

8. Making Connections (EM)

The team linked stakeholder needs with materials properties and considered long-term system implications.

9. Creating Value (EM)

The team focused on improving usability while maintaining technical performance.

10. Feedback Addressed

Instructor feedback was incorporated into the next iteration cycle and influenced design decisions.

Report 30

1. Group Number and Title: Group 11 – Flexible Solar Encapsulation Layer

2. Date: February 27, 2026

3. HCD Spaces Engaged: Ideate, Prototype, Implement, Understand, Synthesize

4. Activities Completed

- Activity 1: Conducted work related to Implement space including data collection, analysis, concept refinement, or material testing.
- Activity 2: Conducted work related to Ideate space including data collection, analysis, concept refinement, or material testing.
- Activity 3: Conducted work related to Synthesize space including data collection, analysis, concept refinement, or material testing.
- Activity 4: Conducted work related to Synthesize space including data collection, analysis, concept refinement, or material testing.

5. Expanded Description

The team worked on assigned tasks and made incremental progress. Limited stakeholder engagement occurred. Decisions were primarily based on technical feasibility.

6. Outstanding Challenges

- Clarify cost constraints.
- Increase stakeholder validation.

7. Curiosity Development (EM)

The team asked clarification questions and followed provided guidance.

8. Making Connections (EM)

Connections were primarily within course materials.

9. Creating Value (EM)

Progress contributed to completing project milestones.

10. Feedback Addressed

Instructor feedback was incorporated into the next iteration cycle and influenced design decisions.

Report 31

1. Group Number and Title: Group 9 – Lightweight Aerospace Alloy Redesign

2. Date: February 23, 2026

3. HCD Spaces Engaged: Synthesize, Prototype, Ideate

4. Activities Completed

- Activity 1: Conducted work related to Ideate space including data collection, analysis, concept refinement, or material testing.
- Activity 2: Conducted work related to Ideate space including data collection, analysis, concept refinement, or material testing.
- Activity 3: Conducted work related to Synthesize space including data collection, analysis, concept refinement, or material testing.
- Activity 4: Conducted work related to Synthesize space including data collection, analysis, concept refinement, or material testing.
- Activity 5: Conducted work related to Synthesize space including data collection, analysis, concept refinement, or material testing.

5. Expanded Description

The team worked on assigned tasks and made incremental progress. Limited stakeholder engagement occurred. Decisions were primarily based on technical feasibility.

6. Outstanding Challenges

- Clarify cost constraints.
- Address manufacturability risks.
- Increase stakeholder validation.

7. Curiosity Development (EM)

The team asked clarification questions and followed provided guidance.

8. Making Connections (EM)

Connections were primarily within course materials.

9. Creating Value (EM)

Progress contributed to completing project milestones.

10. Feedback Addressed

Instructor feedback was incorporated into the next iteration cycle and influenced design decisions.

Report 32

1. Group Number and Title: Group 11 – Antimicrobial Hospital Surface Coating

2. Date: March 22, 2026

3. HCD Spaces Engaged: Ideate, Understand, Implement, Prototype

4. Activities Completed

- Activity 1: Conducted work related to Understand space including data collection, analysis, concept refinement, or material testing.
- Activity 2: Conducted work related to Prototype space including data collection, analysis, concept refinement, or material testing.
- Activity 3: Conducted work related to Implement space including data collection, analysis, concept refinement, or material testing.

5. Expanded Description

The team balanced technical testing with user considerations. Some stakeholder feedback informed concept refinement. Iterative testing improved performance metrics.

6. Outstanding Challenges

- Clarify cost constraints.
- Increase stakeholder validation.
- Address manufacturability risks.

7. Curiosity Development (EM)

The team explored alternative material configurations and questioned initial assumptions about performance trade-offs.

8. Making Connections (EM)

The team linked stakeholder needs with materials properties and considered long-term system implications.

9. Creating Value (EM)

The team focused on improving usability while maintaining technical performance.

10. Feedback Addressed

Instructor feedback was incorporated into the next iteration cycle and influenced design decisions.

Report 33

1. Group Number and Title: Group 23 – Recyclable Polymer Blend Development

2. Date: February 13, 2026

3. HCD Spaces Engaged: Ideate, Implement, Synthesize, Understand

4. Activities Completed

- Activity 1: Conducted work related to Implement space including data collection, analysis, concept refinement, or material testing.
- Activity 2: Conducted work related to Implement space including data collection, analysis, concept refinement, or material testing.
- Activity 3: Conducted work related to Understand space including data collection, analysis, concept refinement, or material testing.
- Activity 4: Conducted work related to Understand space including data collection, analysis, concept refinement, or material testing.
- Activity 5: Conducted work related to Synthesize space including data collection, analysis, concept refinement, or material testing.

5. Expanded Description

The team balanced technical testing with user considerations. Some stakeholder feedback informed concept refinement. Iterative testing improved performance metrics.

6. Outstanding Challenges

- Increase stakeholder validation.
- Clarify cost constraints.
- Address manufacturability risks.
- Improve prototype durability.

7. Curiosity Development (EM)

The team explored alternative material configurations and questioned initial assumptions about performance trade-offs.

8. Making Connections (EM)

The team linked stakeholder needs with materials properties and considered long-term system implications.

9. Creating Value (EM)

The team focused on improving usability while maintaining technical performance.

10. Feedback Addressed

Instructor feedback was incorporated into the next iteration cycle and influenced design decisions.

Report 34

1. Group Number and Title: Group 22 – Heat-Resistant Industrial Glove

2. Date: March 25, 2026

3. HCD Spaces Engaged: Prototype, Understand, Synthesize, Ideate, Implement

4. Activities Completed

- Activity 1: Conducted work related to Understand space including data collection, analysis, concept refinement, or material testing.
- Activity 2: Conducted work related to Implement space including data collection, analysis, concept refinement, or material testing.
- Activity 3: Conducted work related to Ideate space including data collection, analysis, concept refinement, or material testing.

5. Expanded Description

The team deeply integrated stakeholder insights with materials testing. Iterative prototyping informed evidence-based decisions. Trade-offs between performance, manufacturability, and user needs were systematically evaluated.

6. Outstanding Challenges

- Increase stakeholder validation.
- Clarify cost constraints.

7. Curiosity Development (EM)

The team demonstrated inquisitiveness, contrarian thinking, experimentation, and comfort with ambiguity while iterating multiple prototype pathways.

8. Making Connections (EM)

The team applied systems thinking, creativity, and strategic risk management to integrate user insights, engineering constraints, and scalability considerations.

9. Creating Value (EM)

The team demonstrated customer-centric thinking, value awareness, persistence, and continuous improvement while refining prototypes to maximize user impact.

10. Feedback Addressed

Instructor feedback was incorporated into the next iteration cycle and influenced design decisions.

Report 35

1. Group Number and Title: Group 5 – Smart Bio-Composite Packaging

2. Date: March 27, 2026

3. HCD Spaces Engaged: Ideate, Understand, Prototype, Implement

4. Activities Completed

- Activity 1: Conducted work related to Implement space including data collection, analysis, concept refinement, or material testing.
- Activity 2: Conducted work related to Understand space including data collection, analysis, concept refinement, or material testing.
- Activity 3: Conducted work related to Implement space including data collection, analysis, concept refinement, or material testing.
- Activity 4: Conducted work related to Prototype space including data collection, analysis, concept refinement, or material testing.
- Activity 5: Conducted work related to Implement space including data collection, analysis, concept refinement, or material testing.

5. Expanded Description

The team balanced technical testing with user considerations. Some stakeholder feedback informed concept refinement. Iterative testing improved performance metrics.

6. Outstanding Challenges

- Increase stakeholder validation.
- Address manufacturability risks.
- Clarify cost constraints.
- Improve prototype durability.

7. Curiosity Development (EM)

The team explored alternative material configurations and questioned initial assumptions about performance trade-offs.

8. Making Connections (EM)

The team linked stakeholder needs with materials properties and considered long-term system implications.

9. Creating Value (EM)

The team focused on improving usability while maintaining technical performance.

10. Feedback Addressed

Instructor feedback was incorporated into the next iteration cycle and influenced design decisions.

Report 36

1. Group Number and Title: Group 22 – Low-Cost Water Filtration Membrane

2. Date: March 08, 2026

3. HCD Spaces Engaged: Ideate, Synthesize, Implement, Understand, Prototype

4. Activities Completed

- Activity 1: Conducted work related to Understand space including data collection, analysis, concept refinement, or material testing.
- Activity 2: Conducted work related to Understand space including data collection, analysis, concept refinement, or material testing.
- Activity 3: Conducted work related to Ideate space including data collection, analysis, concept refinement, or material testing.

5. Expanded Description

The team deeply integrated stakeholder insights with materials testing. Iterative prototyping informed evidence-based decisions. Trade-offs between performance, manufacturability, and user needs were systematically evaluated.

6. Outstanding Challenges

- Clarify cost constraints.
- Need additional testing data.

7. Curiosity Development (EM)

The team demonstrated inquisitiveness, contrarian thinking, experimentation, and comfort with ambiguity while iterating multiple prototype pathways.

8. Making Connections (EM)

The team applied systems thinking, creativity, and strategic risk management to integrate user insights, engineering constraints, and scalability considerations.

9. Creating Value (EM)

The team demonstrated customer-centric thinking, value awareness, persistence, and continuous improvement while refining prototypes to maximize user impact.

10. Feedback Addressed

Instructor feedback was incorporated into the next iteration cycle and influenced design decisions.

Report 37

1. Group Number and Title: Group 1 – Phase-Change Cooling Vest

2. Date: February 14, 2026

3. HCD Spaces Engaged: Synthesize, Prototype, Ideate

4. Activities Completed

- Activity 1: Conducted work related to Synthesize space including data collection, analysis, concept refinement, or material testing.
- Activity 2: Conducted work related to Ideate space including data collection, analysis, concept refinement, or material testing.
- Activity 3: Conducted work related to Synthesize space including data collection, analysis, concept refinement, or material testing.

5. Expanded Description

The team worked on assigned tasks and made incremental progress. Limited stakeholder engagement occurred. Decisions were primarily based on technical feasibility.

6. Outstanding Challenges

- Need additional testing data.
- Improve prototype durability.
- Clarify cost constraints.
- Increase stakeholder validation.

7. Curiosity Development (EM)

The team asked clarification questions and followed provided guidance.

8. Making Connections (EM)

Connections were primarily within course materials.

9. Creating Value (EM)

Progress contributed to completing project milestones.

10. Feedback Addressed

Instructor feedback was incorporated into the next iteration cycle and influenced design decisions.

Report 38

1. Group Number and Title: Group 19 – Lightweight Aerospace Alloy Redesign

2. Date: March 14, 2026

3. HCD Spaces Engaged: Prototype, Synthesize

4. Activities Completed

- Activity 1: Conducted work related to Prototype space including data collection, analysis, concept refinement, or material testing.
- Activity 2: Conducted work related to Prototype space including data collection, analysis, concept refinement, or material testing.
- Activity 3: Conducted work related to Prototype space including data collection, analysis, concept refinement, or material testing.
- Activity 4: Conducted work related to Synthesize space including data collection, analysis, concept refinement, or material testing.
- Activity 5: Conducted work related to Prototype space including data collection, analysis, concept refinement, or material testing.
- Activity 6: Conducted work related to Synthesize space including data collection, analysis, concept refinement, or material testing.

5. Expanded Description

The team balanced technical testing with user considerations. Some stakeholder feedback informed concept refinement. Iterative testing improved performance metrics.

6. Outstanding Challenges

- Clarify cost constraints.
- Increase stakeholder validation.
- Need additional testing data.

7. Curiosity Development (EM)

The team explored alternative material configurations and questioned initial assumptions about performance trade-offs.

8. Making Connections (EM)

The team linked stakeholder needs with materials properties and considered long-term system implications.

9. Creating Value (EM)

The team focused on improving usability while maintaining technical performance.

10. Feedback Addressed

Instructor feedback was incorporated into the next iteration cycle and influenced design decisions.

Report 39

1. Group Number and Title: Group 16 – Heat-Resistant Industrial Glove

2. Date: February 15, 2026

3. HCD Spaces Engaged: Prototype, Synthesize, Ideate, Implement, Understand

4. Activities Completed

- Activity 1: Conducted work related to Synthesize space including data collection, analysis, concept refinement, or material testing.
- Activity 2: Conducted work related to Understand space including data collection, analysis, concept refinement, or material testing.
- Activity 3: Conducted work related to Prototype space including data collection, analysis, concept refinement, or material testing.

5. Expanded Description

The team worked on assigned tasks and made incremental progress. Limited stakeholder engagement occurred. Decisions were primarily based on technical feasibility.

6. Outstanding Challenges

- Clarify cost constraints.
- Improve prototype durability.

7. Curiosity Development (EM)

The team asked clarification questions and followed provided guidance.

8. Making Connections (EM)

Connections were primarily within course materials.

9. Creating Value (EM)

Progress contributed to completing project milestones.

10. Feedback Addressed

Instructor feedback was incorporated into the next iteration cycle and influenced design decisions.

Report 40

1. Group Number and Title: Group 23 – Lightweight Aerospace Alloy Redesign

2. Date: March 19, 2026

3. HCD Spaces Engaged: Synthesize, Prototype

4. Activities Completed

- Activity 1: Conducted work related to Synthesize space including data collection, analysis, concept refinement, or material testing.
- Activity 2: Conducted work related to Prototype space including data collection, analysis, concept refinement, or material testing.
- Activity 3: Conducted work related to Prototype space including data collection, analysis, concept refinement, or material testing.
- Activity 4: Conducted work related to Synthesize space including data collection, analysis, concept refinement, or material testing.

5. Expanded Description

The team worked on assigned tasks and made incremental progress. Limited stakeholder engagement occurred. Decisions were primarily based on technical feasibility.

6. Outstanding Challenges

- Improve prototype durability.
- Increase stakeholder validation.

7. Curiosity Development (EM)

The team asked clarification questions and followed provided guidance.

8. Making Connections (EM)

Connections were primarily within course materials.

9. Creating Value (EM)

Progress contributed to completing project milestones.

10. Feedback Addressed

Instructor feedback was incorporated into the next iteration cycle and influenced design decisions.

Report 41

1. Group Number and Title: Group 8 – Heat-Resistant Industrial Glove

2. Date: February 18, 2026

3. HCD Spaces Engaged: Understand, Prototype, Implement, Ideate, Synthesize

4. Activities Completed

- Activity 1: Conducted work related to Prototype space including data collection, analysis, concept refinement, or material testing.
- Activity 2: Conducted work related to Understand space including data collection, analysis, concept refinement, or material testing.
- Activity 3: Conducted work related to Prototype space including data collection, analysis, concept refinement, or material testing.
- Activity 4: Conducted work related to Understand space including data collection, analysis, concept refinement, or material testing.

5. Expanded Description

The team balanced technical testing with user considerations. Some stakeholder feedback informed concept refinement. Iterative testing improved performance metrics.

6. Outstanding Challenges

- Need additional testing data.
- Increase stakeholder validation.
- Address manufacturability risks.
- Clarify cost constraints.

7. Curiosity Development (EM)

The team explored alternative material configurations and questioned initial assumptions about performance trade-offs.

8. Making Connections (EM)

The team linked stakeholder needs with materials properties and considered long-term system implications.

9. Creating Value (EM)

The team focused on improving usability while maintaining technical performance.

10. Feedback Addressed

Instructor feedback was incorporated into the next iteration cycle and influenced design decisions.

Report 42

1. Group Number and Title: Group 17 – Sustainable Concrete Additive

2. Date: March 02, 2026

3. HCD Spaces Engaged: Prototype, Ideate, Synthesize, Implement

4. Activities Completed

- Activity 1: Conducted work related to Synthesize space including data collection, analysis, concept refinement, or material testing.
- Activity 2: Conducted work related to Synthesize space including data collection, analysis, concept refinement, or material testing.
- Activity 3: Conducted work related to Prototype space including data collection, analysis, concept refinement, or material testing.
- Activity 4: Conducted work related to Synthesize space including data collection, analysis, concept refinement, or material testing.
- Activity 5: Conducted work related to Ideate space including data collection, analysis, concept refinement, or material testing.

5. Expanded Description

The team worked on assigned tasks and made incremental progress. Limited stakeholder engagement occurred. Decisions were primarily based on technical feasibility.

6. Outstanding Challenges

- Address manufacturability risks.
- Increase stakeholder validation.
- Clarify cost constraints.

7. Curiosity Development (EM)

The team asked clarification questions and followed provided guidance.

8. Making Connections (EM)

Connections were primarily within course materials.

9. Creating Value (EM)

Progress contributed to completing project milestones.

10. Feedback Addressed

Instructor feedback was incorporated into the next iteration cycle and influenced design decisions.

Report 43

1. Group Number and Title: Group 4 – Wearable Thermal Regulation System

2. Date: March 04, 2026

3. HCD Spaces Engaged: Implement, Prototype, Ideate, Synthesize, Understand

4. Activities Completed

- Activity 1: Conducted work related to Ideate space including data collection, analysis, concept refinement, or material testing.
- Activity 2: Conducted work related to Ideate space including data collection, analysis, concept refinement, or material testing.
- Activity 3: Conducted work related to Implement space including data collection, analysis, concept refinement, or material testing.
- Activity 4: Conducted work related to Implement space including data collection, analysis, concept refinement, or material testing.

5. Expanded Description

The team worked on assigned tasks and made incremental progress. Limited stakeholder engagement occurred. Decisions were primarily based on technical feasibility.

6. Outstanding Challenges

- Improve prototype durability.
- Need additional testing data.

7. Curiosity Development (EM)

The team asked clarification questions and followed provided guidance.

8. Making Connections (EM)

Connections were primarily within course materials.

9. Creating Value (EM)

Progress contributed to completing project milestones.

10. Feedback Addressed

Instructor feedback was incorporated into the next iteration cycle and influenced design decisions.

Report 44

1. Group Number and Title: Group 24 – Flexible Solar Encapsulation Layer

2. Date: February 12, 2026

3. HCD Spaces Engaged: Prototype, Implement

4. Activities Completed

- Activity 1: Conducted work related to Implement space including data collection, analysis, concept refinement, or material testing.
- Activity 2: Conducted work related to Implement space including data collection, analysis, concept refinement, or material testing.
- Activity 3: Conducted work related to Prototype space including data collection, analysis, concept refinement, or material testing.
- Activity 4: Conducted work related to Implement space including data collection, analysis, concept refinement, or material testing.
- Activity 5: Conducted work related to Prototype space including data collection, analysis, concept refinement, or material testing.

5. Expanded Description

The team balanced technical testing with user considerations. Some stakeholder feedback informed concept refinement. Iterative testing improved performance metrics.

6. Outstanding Challenges

- Address manufacturability risks.
- Need additional testing data.
- Improve prototype durability.
- Increase stakeholder validation.

7. Curiosity Development (EM)

The team explored alternative material configurations and questioned initial assumptions about performance trade-offs.

8. Making Connections (EM)

The team linked stakeholder needs with materials properties and considered long-term system implications.

9. Creating Value (EM)

The team focused on improving usability while maintaining technical performance.

10. Feedback Addressed

Instructor feedback was incorporated into the next iteration cycle and influenced design decisions.

Report 45

1. Group Number and Title: Group 23 – Phase-Change Cooling Vest

2. Date: February 14, 2026

3. HCD Spaces Engaged: Synthesize, Ideate, Implement

4. Activities Completed

- Activity 1: Conducted work related to Synthesize space including data collection, analysis, concept refinement, or material testing.
- Activity 2: Conducted work related to Implement space including data collection, analysis, concept refinement, or material testing.
- Activity 3: Conducted work related to Implement space including data collection, analysis, concept refinement, or material testing.

5. Expanded Description

The team balanced technical testing with user considerations. Some stakeholder feedback informed concept refinement. Iterative testing improved performance metrics.

6. Outstanding Challenges

- Increase stakeholder validation.
- Improve prototype durability.
- Clarify cost constraints.

7. Curiosity Development (EM)

The team explored alternative material configurations and questioned initial assumptions about performance trade-offs.

8. Making Connections (EM)

The team linked stakeholder needs with materials properties and considered long-term system implications.

9. Creating Value (EM)

The team focused on improving usability while maintaining technical performance.

10. Feedback Addressed

Instructor feedback was incorporated into the next iteration cycle and influenced design decisions.

Report 46

1. Group Number and Title: Group 7 – Antimicrobial Hospital Surface Coating

2. Date: March 11, 2026

3. HCD Spaces Engaged: Understand, Prototype

4. Activities Completed

- Activity 1: Conducted work related to Understand space including data collection, analysis, concept refinement, or material testing.
- Activity 2: Conducted work related to Understand space including data collection, analysis, concept refinement, or material testing.
- Activity 3: Conducted work related to Prototype space including data collection, analysis, concept refinement, or material testing.

5. Expanded Description

The team deeply integrated stakeholder insights with materials testing. Iterative prototyping informed evidence-based decisions. Trade-offs between performance, manufacturability, and user needs were systematically evaluated.

6. Outstanding Challenges

- Clarify cost constraints.
- Improve prototype durability.
- Address manufacturability risks.
- Need additional testing data.

7. Curiosity Development (EM)

The team demonstrated inquisitiveness, contrarian thinking, experimentation, and comfort with ambiguity while iterating multiple prototype pathways.

8. Making Connections (EM)

The team applied systems thinking, creativity, and strategic risk management to integrate user insights, engineering constraints, and scalability considerations.

9. Creating Value (EM)

The team demonstrated customer-centric thinking, value awareness, persistence, and continuous improvement while refining prototypes to maximize user impact.

10. Feedback Addressed

Instructor feedback was incorporated into the next iteration cycle and influenced design decisions.

Report 47

1. Group Number and Title: Group 19 – Sustainable Concrete Additive

2. Date: February 27, 2026

3. HCD Spaces Engaged: Implement, Ideate

4. Activities Completed

- Activity 1: Conducted work related to Ideate space including data collection, analysis, concept refinement, or material testing.
- Activity 2: Conducted work related to Implement space including data collection, analysis, concept refinement, or material testing.
- Activity 3: Conducted work related to Ideate space including data collection, analysis, concept refinement, or material testing.
- Activity 4: Conducted work related to Implement space including data collection, analysis, concept refinement, or material testing.

5. Expanded Description

The team balanced technical testing with user considerations. Some stakeholder feedback informed concept refinement. Iterative testing improved performance metrics.

6. Outstanding Challenges

- Need additional testing data.
- Clarify cost constraints.

7. Curiosity Development (EM)

The team explored alternative material configurations and questioned initial assumptions about performance trade-offs.

8. Making Connections (EM)

The team linked stakeholder needs with materials properties and considered long-term system implications.

9. Creating Value (EM)

The team focused on improving usability while maintaining technical performance.

10. Feedback Addressed

Instructor feedback was incorporated into the next iteration cycle and influenced design decisions.

Report 48

1. Group Number and Title: Group 11 – Antimicrobial Hospital Surface Coating

2. Date: February 08, 2026

3. HCD Spaces Engaged: Implement, Prototype, Ideate, Understand, Synthesize

4. Activities Completed

- Activity 1: Conducted work related to Implement space including data collection, analysis, concept refinement, or material testing.
- Activity 2: Conducted work related to Understand space including data collection, analysis, concept refinement, or material testing.
- Activity 3: Conducted work related to Synthesize space including data collection, analysis, concept refinement, or material testing.

5. Expanded Description

The team balanced technical testing with user considerations. Some stakeholder feedback informed concept refinement. Iterative testing improved performance metrics.

6. Outstanding Challenges

- Need additional testing data.
- Address manufacturability risks.

7. Curiosity Development (EM)

The team explored alternative material configurations and questioned initial assumptions about performance trade-offs.

8. Making Connections (EM)

The team linked stakeholder needs with materials properties and considered long-term system implications.

9. Creating Value (EM)

The team focused on improving usability while maintaining technical performance.

10. Feedback Addressed

Instructor feedback was incorporated into the next iteration cycle and influenced design decisions.

Report 49

1. Group Number and Title: Group 6 – Heat-Resistant Industrial Glove

2. Date: February 25, 2026

3. HCD Spaces Engaged: Implement, Synthesize, Ideate, Prototype

4. Activities Completed

- Activity 1: Conducted work related to Synthesize space including data collection, analysis, concept refinement, or material testing.
- Activity 2: Conducted work related to Ideate space including data collection, analysis, concept refinement, or material testing.
- Activity 3: Conducted work related to Prototype space including data collection, analysis, concept refinement, or material testing.
- Activity 4: Conducted work related to Ideate space including data collection, analysis, concept refinement, or material testing.
- Activity 5: Conducted work related to Prototype space including data collection, analysis, concept refinement, or material testing.
- Activity 6: Conducted work related to Implement space including data collection, analysis, concept refinement, or material testing.

5. Expanded Description

The team worked on assigned tasks and made incremental progress. Limited stakeholder engagement occurred. Decisions were primarily based on technical feasibility.

6. Outstanding Challenges

- Address manufacturability risks.
- Need additional testing data.
- Increase stakeholder validation.

7. Curiosity Development (EM)

The team asked clarification questions and followed provided guidance.

8. Making Connections (EM)

Connections were primarily within course materials.

9. Creating Value (EM)

Progress contributed to completing project milestones.

10. Feedback Addressed

Instructor feedback was incorporated into the next iteration cycle and influenced design decisions.

Report 50

1. Group Number and Title: Group 3 – Phase-Change Cooling Vest

2. Date: February 06, 2026

3. HCD Spaces Engaged: Ideate, Synthesize, Prototype, Understand

4. Activities Completed

- Activity 1: Conducted work related to Understand space including data collection, analysis, concept refinement, or material testing.
- Activity 2: Conducted work related to Understand space including data collection, analysis, concept refinement, or material testing.
- Activity 3: Conducted work related to Ideate space including data collection, analysis, concept refinement, or material testing.
- Activity 4: Conducted work related to Ideate space including data collection, analysis, concept refinement, or material testing.

5. Expanded Description

The team worked on assigned tasks and made incremental progress. Limited stakeholder engagement occurred. Decisions were primarily based on technical feasibility.

6. Outstanding Challenges

- Need additional testing data.
- Clarify cost constraints.

7. Curiosity Development (EM)

The team asked clarification questions and followed provided guidance.

8. Making Connections (EM)

Connections were primarily within course materials.

9. Creating Value (EM)

Progress contributed to completing project milestones.

10. Feedback Addressed

Instructor feedback was incorporated into the next iteration cycle and influenced design decisions.

Report 51

1. Group Number and Title: Group 10 – Recyclable Polymer Blend Development

2. Date: February 03, 2026

3. HCD Spaces Engaged: Synthesize, Understand, Implement, Prototype, Ideate

4. Activities Completed

- Activity 1: Conducted work related to Implement space including data collection, analysis, concept refinement, or material testing.
- Activity 2: Conducted work related to Ideate space including data collection, analysis, concept refinement, or material testing.
- Activity 3: Conducted work related to Synthesize space including data collection, analysis, concept refinement, or material testing.
- Activity 4: Conducted work related to Synthesize space including data collection, analysis, concept refinement, or material testing.

5. Expanded Description

The team deeply integrated stakeholder insights with materials testing. Iterative prototyping informed evidence-based decisions. Trade-offs between performance, manufacturability, and user needs were systematically evaluated.

6. Outstanding Challenges

- Need additional testing data.
- Clarify cost constraints.
- Address manufacturability risks.

7. Curiosity Development (EM)

The team demonstrated inquisitiveness, contrarian thinking, experimentation, and comfort with ambiguity while iterating multiple prototype pathways.

8. Making Connections (EM)

The team applied systems thinking, creativity, and strategic risk management to integrate user insights, engineering constraints, and scalability considerations.

9. Creating Value (EM)

The team demonstrated customer-centric thinking, value awareness, persistence, and continuous improvement while refining prototypes to maximize user impact.

10. Feedback Addressed

Instructor feedback was incorporated into the next iteration cycle and influenced design decisions.

Report 52

1. Group Number and Title: Group 2 – Recyclable Polymer Blend Development

2. Date: March 11, 2026

3. HCD Spaces Engaged: Synthesize, Ideate

4. Activities Completed

- Activity 1: Conducted work related to Synthesize space including data collection, analysis, concept refinement, or material testing.
- Activity 2: Conducted work related to Ideate space including data collection, analysis, concept refinement, or material testing.
- Activity 3: Conducted work related to Ideate space including data collection, analysis, concept refinement, or material testing.

5. Expanded Description

The team deeply integrated stakeholder insights with materials testing. Iterative prototyping informed evidence-based decisions. Trade-offs between performance, manufacturability, and user needs were systematically evaluated.

6. Outstanding Challenges

- Improve prototype durability.
- Address manufacturability risks.
- Need additional testing data.

7. Curiosity Development (EM)

The team demonstrated inquisitiveness, contrarian thinking, experimentation, and comfort with ambiguity while iterating multiple prototype pathways.

8. Making Connections (EM)

The team applied systems thinking, creativity, and strategic risk management to integrate user insights, engineering constraints, and scalability considerations.

9. Creating Value (EM)

The team demonstrated customer-centric thinking, value awareness, persistence, and continuous improvement while refining prototypes to maximize user impact.

10. Feedback Addressed

Instructor feedback was incorporated into the next iteration cycle and influenced design decisions.

Report 53

1. Group Number and Title: Group 10 – Low-Cost Water Filtration Membrane

2. Date: March 15, 2026

3. HCD Spaces Engaged: Prototype, Synthesize, Ideate

4. Activities Completed

- Activity 1: Conducted work related to Ideate space including data collection, analysis, concept refinement, or material testing.
- Activity 2: Conducted work related to Synthesize space including data collection, analysis, concept refinement, or material testing.
- Activity 3: Conducted work related to Ideate space including data collection, analysis, concept refinement, or material testing.

5. Expanded Description

The team deeply integrated stakeholder insights with materials testing. Iterative prototyping informed evidence-based decisions. Trade-offs between performance, manufacturability, and user needs were systematically evaluated.

6. Outstanding Challenges

- Increase stakeholder validation.
- Improve prototype durability.

7. Curiosity Development (EM)

The team demonstrated inquisitiveness, contrarian thinking, experimentation, and comfort with ambiguity while iterating multiple prototype pathways.

8. Making Connections (EM)

The team applied systems thinking, creativity, and strategic risk management to integrate user insights, engineering constraints, and scalability considerations.

9. Creating Value (EM)

The team demonstrated customer-centric thinking, value awareness, persistence, and continuous improvement while refining prototypes to maximize user impact.

10. Feedback Addressed

Instructor feedback was incorporated into the next iteration cycle and influenced design decisions.

Report 54

1. Group Number and Title: Group 6 – Sustainable Concrete Additive

2. Date: February 25, 2026

3. HCD Spaces Engaged: Synthesize, Prototype, Ideate

4. Activities Completed

- Activity 1: Conducted work related to Prototype space including data collection, analysis, concept refinement, or material testing.
- Activity 2: Conducted work related to Prototype space including data collection, analysis, concept refinement, or material testing.
- Activity 3: Conducted work related to Prototype space including data collection, analysis, concept refinement, or material testing.
- Activity 4: Conducted work related to Synthesize space including data collection, analysis, concept refinement, or material testing.

5. Expanded Description

The team deeply integrated stakeholder insights with materials testing. Iterative prototyping informed evidence-based decisions. Trade-offs between performance, manufacturability, and user needs were systematically evaluated.

6. Outstanding Challenges

- Improve prototype durability.
- Increase stakeholder validation.

7. Curiosity Development (EM)

The team demonstrated inquisitiveness, contrarian thinking, experimentation, and comfort with ambiguity while iterating multiple prototype pathways.

8. Making Connections (EM)

The team applied systems thinking, creativity, and strategic risk management to integrate user insights, engineering constraints, and scalability considerations.

9. Creating Value (EM)

The team demonstrated customer-centric thinking, value awareness, persistence, and continuous improvement while refining prototypes to maximize user impact.

10. Feedback Addressed

Instructor feedback was incorporated into the next iteration cycle and influenced design decisions.

Report 55

1. Group Number and Title: Group 7 – Antimicrobial Hospital Surface Coating

2. Date: February 24, 2026

3. HCD Spaces Engaged: Synthesize, Understand, Prototype

4. Activities Completed

- Activity 1: Conducted work related to Understand space including data collection, analysis, concept refinement, or material testing.
- Activity 2: Conducted work related to Understand space including data collection, analysis, concept refinement, or material testing.
- Activity 3: Conducted work related to Prototype space including data collection, analysis, concept refinement, or material testing.
- Activity 4: Conducted work related to Prototype space including data collection, analysis, concept refinement, or material testing.
- Activity 5: Conducted work related to Prototype space including data collection, analysis, concept refinement, or material testing.

5. Expanded Description

The team balanced technical testing with user considerations. Some stakeholder feedback informed concept refinement. Iterative testing improved performance metrics.

6. Outstanding Challenges

- Clarify cost constraints.
- Increase stakeholder validation.

7. Curiosity Development (EM)

The team explored alternative material configurations and questioned initial assumptions about performance trade-offs.

8. Making Connections (EM)

The team linked stakeholder needs with materials properties and considered long-term system implications.

9. Creating Value (EM)

The team focused on improving usability while maintaining technical performance.

10. Feedback Addressed

Instructor feedback was incorporated into the next iteration cycle and influenced design decisions.

Report 56

1. Group Number and Title: Group 7 – Phase-Change Cooling Vest

2. Date: March 11, 2026

3. HCD Spaces Engaged: Synthesize, Ideate, Understand

4. Activities Completed

- Activity 1: Conducted work related to Ideate space including data collection, analysis, concept refinement, or material testing.
- Activity 2: Conducted work related to Understand space including data collection, analysis, concept refinement, or material testing.
- Activity 3: Conducted work related to Ideate space including data collection, analysis, concept refinement, or material testing.

5. Expanded Description

The team worked on assigned tasks and made incremental progress. Limited stakeholder engagement occurred. Decisions were primarily based on technical feasibility.

6. Outstanding Challenges

- Improve prototype durability.
- Clarify cost constraints.
- Need additional testing data.

7. Curiosity Development (EM)

The team asked clarification questions and followed provided guidance.

8. Making Connections (EM)

Connections were primarily within course materials.

9. Creating Value (EM)

Progress contributed to completing project milestones.

10. Feedback Addressed

Instructor feedback was incorporated into the next iteration cycle and influenced design decisions.

Report 57

1. Group Number and Title: Group 17 – Heat-Resistant Industrial Glove

2. Date: February 22, 2026

3. HCD Spaces Engaged: Synthesize, Understand, Prototype, Implement, Ideate

4. Activities Completed

- Activity 1: Conducted work related to Implement space including data collection, analysis, concept refinement, or material testing.
- Activity 2: Conducted work related to Ideate space including data collection, analysis, concept refinement, or material testing.
- Activity 3: Conducted work related to Ideate space including data collection, analysis, concept refinement, or material testing.
- Activity 4: Conducted work related to Ideate space including data collection, analysis, concept refinement, or material testing.

5. Expanded Description

The team deeply integrated stakeholder insights with materials testing. Iterative prototyping informed evidence-based decisions. Trade-offs between performance, manufacturability, and user needs were systematically evaluated.

6. Outstanding Challenges

- Address manufacturability risks.
- Need additional testing data.

7. Curiosity Development (EM)

The team demonstrated inquisitiveness, contrarian thinking, experimentation, and comfort with ambiguity while iterating multiple prototype pathways.

8. Making Connections (EM)

The team applied systems thinking, creativity, and strategic risk management to integrate user insights, engineering constraints, and scalability considerations.

9. Creating Value (EM)

The team demonstrated customer-centric thinking, value awareness, persistence, and continuous improvement while refining prototypes to maximize user impact.

10. Feedback Addressed

Instructor feedback was incorporated into the next iteration cycle and influenced design decisions.

Report 58

1. Group Number and Title: Group 12 – Flexible Solar Encapsulation Layer

2. Date: February 25, 2026

3. HCD Spaces Engaged: Prototype, Ideate, Implement, Synthesize, Understand

4. Activities Completed

- Activity 1: Conducted work related to Understand space including data collection, analysis, concept refinement, or material testing.
- Activity 2: Conducted work related to Implement space including data collection, analysis, concept refinement, or material testing.
- Activity 3: Conducted work related to Implement space including data collection, analysis, concept refinement, or material testing.

5. Expanded Description

The team worked on assigned tasks and made incremental progress. Limited stakeholder engagement occurred. Decisions were primarily based on technical feasibility.

6. Outstanding Challenges

- Clarify cost constraints.
- Improve prototype durability.
- Need additional testing data.
- Address manufacturability risks.

7. Curiosity Development (EM)

The team asked clarification questions and followed provided guidance.

8. Making Connections (EM)

Connections were primarily within course materials.

9. Creating Value (EM)

Progress contributed to completing project milestones.

10. Feedback Addressed

Instructor feedback was incorporated into the next iteration cycle and influenced design decisions.

Report 59

1. Group Number and Title: Group 2 – Low-Cost Water Filtration Membrane

2. Date: February 27, 2026

3. HCD Spaces Engaged: Ideate, Implement, Prototype, Understand, Synthesize

4. Activities Completed

- Activity 1: Conducted work related to Prototype space including data collection, analysis, concept refinement, or material testing.
- Activity 2: Conducted work related to Synthesize space including data collection, analysis, concept refinement, or material testing.
- Activity 3: Conducted work related to Understand space including data collection, analysis, concept refinement, or material testing.
- Activity 4: Conducted work related to Ideate space including data collection, analysis, concept refinement, or material testing.

5. Expanded Description

The team deeply integrated stakeholder insights with materials testing. Iterative prototyping informed evidence-based decisions. Trade-offs between performance, manufacturability, and user needs were systematically evaluated.

6. Outstanding Challenges

- Need additional testing data.
- Increase stakeholder validation.
- Clarify cost constraints.

7. Curiosity Development (EM)

The team demonstrated inquisitiveness, contrarian thinking, experimentation, and comfort with ambiguity while iterating multiple prototype pathways.

8. Making Connections (EM)

The team applied systems thinking, creativity, and strategic risk management to integrate user insights, engineering constraints, and scalability considerations.

9. Creating Value (EM)

The team demonstrated customer-centric thinking, value awareness, persistence, and continuous improvement while refining prototypes to maximize user impact.

10. Feedback Addressed

Instructor feedback was incorporated into the next iteration cycle and influenced design decisions.

Report 60

1. Group Number and Title: Group 9 – Antimicrobial Hospital Surface Coating

2. Date: March 05, 2026

3. HCD Spaces Engaged: Prototype, Understand, Ideate, Implement

4. Activities Completed

- Activity 1: Conducted work related to Implement space including data collection, analysis, concept refinement, or material testing.
- Activity 2: Conducted work related to Implement space including data collection, analysis, concept refinement, or material testing.
- Activity 3: Conducted work related to Prototype space including data collection, analysis, concept refinement, or material testing.

5. Expanded Description

The team worked on assigned tasks and made incremental progress. Limited stakeholder engagement occurred. Decisions were primarily based on technical feasibility.

6. Outstanding Challenges

- Need additional testing data.
- Improve prototype durability.
- Address manufacturability risks.

7. Curiosity Development (EM)

The team asked clarification questions and followed provided guidance.

8. Making Connections (EM)

Connections were primarily within course materials.

9. Creating Value (EM)

Progress contributed to completing project milestones.

10. Feedback Addressed

Instructor feedback was incorporated into the next iteration cycle and influenced design decisions.

Report 61

1. Group Number and Title: Group 8 – Flexible Solar Encapsulation Layer

2. Date: February 18, 2026

3. HCD Spaces Engaged: Ideate, Implement, Understand, Prototype, Synthesize

4. Activities Completed

- Activity 1: Conducted work related to Synthesize space including data collection, analysis, concept refinement, or material testing.
- Activity 2: Conducted work related to Ideate space including data collection, analysis, concept refinement, or material testing.
- Activity 3: Conducted work related to Understand space including data collection, analysis, concept refinement, or material testing.
- Activity 4: Conducted work related to Implement space including data collection, analysis, concept refinement, or material testing.
- Activity 5: Conducted work related to Ideate space including data collection, analysis, concept refinement, or material testing.

5. Expanded Description

The team balanced technical testing with user considerations. Some stakeholder feedback informed concept refinement. Iterative testing improved performance metrics.

6. Outstanding Challenges

- Clarify cost constraints.
- Increase stakeholder validation.
- Need additional testing data.
- Improve prototype durability.

7. Curiosity Development (EM)

The team explored alternative material configurations and questioned initial assumptions about performance trade-offs.

8. Making Connections (EM)

The team linked stakeholder needs with materials properties and considered long-term system implications.

9. Creating Value (EM)

The team focused on improving usability while maintaining technical performance.

10. Feedback Addressed

Instructor feedback was incorporated into the next iteration cycle and influenced design decisions.

Report 62

1. Group Number and Title: Group 8 – Flexible Solar Encapsulation Layer

2. Date: March 10, 2026

3. HCD Spaces Engaged: Ideate, Implement, Understand

4. Activities Completed

- Activity 1: Conducted work related to Implement space including data collection, analysis, concept refinement, or material testing.
- Activity 2: Conducted work related to Understand space including data collection, analysis, concept refinement, or material testing.
- Activity 3: Conducted work related to Ideate space including data collection, analysis, concept refinement, or material testing.
- Activity 4: Conducted work related to Ideate space including data collection, analysis, concept refinement, or material testing.
- Activity 5: Conducted work related to Understand space including data collection, analysis, concept refinement, or material testing.
- Activity 6: Conducted work related to Implement space including data collection, analysis, concept refinement, or material testing.

5. Expanded Description

The team balanced technical testing with user considerations. Some stakeholder feedback informed concept refinement. Iterative testing improved performance metrics.

6. Outstanding Challenges

- Need additional testing data.
- Improve prototype durability.

7. Curiosity Development (EM)

The team explored alternative material configurations and questioned initial assumptions about performance trade-offs.

8. Making Connections (EM)

The team linked stakeholder needs with materials properties and considered long-term system implications.

9. Creating Value (EM)

The team focused on improving usability while maintaining technical performance.

10. Feedback Addressed

Instructor feedback was incorporated into the next iteration cycle and influenced design decisions.

Report 63

1. Group Number and Title: Group 6 – Lightweight Aerospace Alloy Redesign

2. Date: February 25, 2026

3. HCD Spaces Engaged: Synthesize, Implement, Understand

4. Activities Completed

- Activity 1: Conducted work related to Implement space including data collection, analysis, concept refinement, or material testing.
- Activity 2: Conducted work related to Synthesize space including data collection, analysis, concept refinement, or material testing.
- Activity 3: Conducted work related to Understand space including data collection, analysis, concept refinement, or material testing.
- Activity 4: Conducted work related to Implement space including data collection, analysis, concept refinement, or material testing.
- Activity 5: Conducted work related to Synthesize space including data collection, analysis, concept refinement, or material testing.
- Activity 6: Conducted work related to Understand space including data collection, analysis, concept refinement, or material testing.

5. Expanded Description

The team worked on assigned tasks and made incremental progress. Limited stakeholder engagement occurred. Decisions were primarily based on technical feasibility.

6. Outstanding Challenges

- Clarify cost constraints.
- Improve prototype durability.

7. Curiosity Development (EM)

The team asked clarification questions and followed provided guidance.

8. Making Connections (EM)

Connections were primarily within course materials.

9. Creating Value (EM)

Progress contributed to completing project milestones.

10. Feedback Addressed

Instructor feedback was incorporated into the next iteration cycle and influenced design decisions.

Report 64

1. Group Number and Title: Group 17 – Phase-Change Cooling Vest

2. Date: March 22, 2026

3. HCD Spaces Engaged: Synthesize, Implement, Understand

4. Activities Completed

- Activity 1: Conducted work related to Synthesize space including data collection, analysis, concept refinement, or material testing.
- Activity 2: Conducted work related to Understand space including data collection, analysis, concept refinement, or material testing.
- Activity 3: Conducted work related to Implement space including data collection, analysis, concept refinement, or material testing.

5. Expanded Description

The team worked on assigned tasks and made incremental progress. Limited stakeholder engagement occurred. Decisions were primarily based on technical feasibility.

6. Outstanding Challenges

- Need additional testing data.
- Improve prototype durability.
- Address manufacturability risks.
- Increase stakeholder validation.

7. Curiosity Development (EM)

The team asked clarification questions and followed provided guidance.

8. Making Connections (EM)

Connections were primarily within course materials.

9. Creating Value (EM)

Progress contributed to completing project milestones.

10. Feedback Addressed

Instructor feedback was incorporated into the next iteration cycle and influenced design decisions.

Report 65

1. Group Number and Title: Group 3 – Wearable Thermal Regulation System

2. Date: March 27, 2026

3. HCD Spaces Engaged: Ideate, Synthesize, Prototype, Implement, Understand

4. Activities Completed

- Activity 1: Conducted work related to Implement space including data collection, analysis, concept refinement, or material testing.
- Activity 2: Conducted work related to Understand space including data collection, analysis, concept refinement, or material testing.
- Activity 3: Conducted work related to Understand space including data collection, analysis, concept refinement, or material testing.
- Activity 4: Conducted work related to Synthesize space including data collection, analysis, concept refinement, or material testing.
- Activity 5: Conducted work related to Synthesize space including data collection, analysis, concept refinement, or material testing.

5. Expanded Description

The team deeply integrated stakeholder insights with materials testing. Iterative prototyping informed evidence-based decisions. Trade-offs between performance, manufacturability, and user needs were systematically evaluated.

6. Outstanding Challenges

- Address manufacturability risks.
- Clarify cost constraints.
- Increase stakeholder validation.

7. Curiosity Development (EM)

The team demonstrated inquisitiveness, contrarian thinking, experimentation, and comfort with ambiguity while iterating multiple prototype pathways.

8. Making Connections (EM)

The team applied systems thinking, creativity, and strategic risk management to integrate user insights, engineering constraints, and scalability considerations.

9. Creating Value (EM)

The team demonstrated customer-centric thinking, value awareness, persistence, and continuous improvement while refining prototypes to maximize user impact.

10. Feedback Addressed

Instructor feedback was incorporated into the next iteration cycle and influenced design decisions.

Report 66

1. Group Number and Title: Group 3 – Antimicrobial Hospital Surface Coating

2. Date: February 13, 2026

3. HCD Spaces Engaged: Ideate, Implement

4. Activities Completed

- Activity 1: Conducted work related to Implement space including data collection, analysis, concept refinement, or material testing.
- Activity 2: Conducted work related to Ideate space including data collection, analysis, concept refinement, or material testing.
- Activity 3: Conducted work related to Implement space including data collection, analysis, concept refinement, or material testing.

5. Expanded Description

The team deeply integrated stakeholder insights with materials testing. Iterative prototyping informed evidence-based decisions. Trade-offs between performance, manufacturability, and user needs were systematically evaluated.

6. Outstanding Challenges

- Clarify cost constraints.
- Need additional testing data.

7. Curiosity Development (EM)

The team demonstrated inquisitiveness, contrarian thinking, experimentation, and comfort with ambiguity while iterating multiple prototype pathways.

8. Making Connections (EM)

The team applied systems thinking, creativity, and strategic risk management to integrate user insights, engineering constraints, and scalability considerations.

9. Creating Value (EM)

The team demonstrated customer-centric thinking, value awareness, persistence, and continuous improvement while refining prototypes to maximize user impact.

10. Feedback Addressed

Instructor feedback was incorporated into the next iteration cycle and influenced design decisions.

Report 67

1. Group Number and Title: Group 17 – Lightweight Aerospace Alloy Redesign

2. Date: February 22, 2026

3. HCD Spaces Engaged: Ideate, Implement, Synthesize, Prototype, Understand

4. Activities Completed

- Activity 1: Conducted work related to Ideate space including data collection, analysis, concept refinement, or material testing.
- Activity 2: Conducted work related to Understand space including data collection, analysis, concept refinement, or material testing.
- Activity 3: Conducted work related to Ideate space including data collection, analysis, concept refinement, or material testing.

5. Expanded Description

The team worked on assigned tasks and made incremental progress. Limited stakeholder engagement occurred. Decisions were primarily based on technical feasibility.

6. Outstanding Challenges

- Clarify cost constraints.
- Improve prototype durability.

7. Curiosity Development (EM)

The team asked clarification questions and followed provided guidance.

8. Making Connections (EM)

Connections were primarily within course materials.

9. Creating Value (EM)

Progress contributed to completing project milestones.

10. Feedback Addressed

Instructor feedback was incorporated into the next iteration cycle and influenced design decisions.

Report 68

1. Group Number and Title: Group 17 – Phase-Change Cooling Vest

2. Date: March 19, 2026

3. HCD Spaces Engaged: Synthesize, Ideate, Implement, Prototype, Understand

4. Activities Completed

- Activity 1: Conducted work related to Implement space including data collection, analysis, concept refinement, or material testing.
- Activity 2: Conducted work related to Ideate space including data collection, analysis, concept refinement, or material testing.
- Activity 3: Conducted work related to Prototype space including data collection, analysis, concept refinement, or material testing.
- Activity 4: Conducted work related to Ideate space including data collection, analysis, concept refinement, or material testing.
- Activity 5: Conducted work related to Understand space including data collection, analysis, concept refinement, or material testing.
- Activity 6: Conducted work related to Synthesize space including data collection, analysis, concept refinement, or material testing.

5. Expanded Description

The team deeply integrated stakeholder insights with materials testing. Iterative prototyping informed evidence-based decisions. Trade-offs between performance, manufacturability, and user needs were systematically evaluated.

6. Outstanding Challenges

- Address manufacturability risks.
- Improve prototype durability.
- Clarify cost constraints.
- Increase stakeholder validation.

7. Curiosity Development (EM)

The team demonstrated inquisitiveness, contrarian thinking, experimentation, and comfort with ambiguity while iterating multiple prototype pathways.

8. Making Connections (EM)

The team applied systems thinking, creativity, and strategic risk management to integrate user insights, engineering constraints, and scalability considerations.

9. Creating Value (EM)

The team demonstrated customer-centric thinking, value awareness, persistence, and continuous improvement while refining prototypes to maximize user impact.

10. Feedback Addressed

Instructor feedback was incorporated into the next iteration cycle and influenced design decisions.

Report 69

1. Group Number and Title: Group 22 – Lightweight Aerospace Alloy Redesign

2. Date: February 08, 2026

3. HCD Spaces Engaged: Implement, Understand

4. Activities Completed

- Activity 1: Conducted work related to Understand space including data collection, analysis, concept refinement, or material testing.
- Activity 2: Conducted work related to Implement space including data collection, analysis, concept refinement, or material testing.
- Activity 3: Conducted work related to Implement space including data collection, analysis, concept refinement, or material testing.
- Activity 4: Conducted work related to Implement space including data collection, analysis, concept refinement, or material testing.
- Activity 5: Conducted work related to Understand space including data collection, analysis, concept refinement, or material testing.

5. Expanded Description

The team balanced technical testing with user considerations. Some stakeholder feedback informed concept refinement. Iterative testing improved performance metrics.

6. Outstanding Challenges

- Address manufacturability risks.
- Increase stakeholder validation.
- Need additional testing data.
- Clarify cost constraints.

7. Curiosity Development (EM)

The team explored alternative material configurations and questioned initial assumptions about performance trade-offs.

8. Making Connections (EM)

The team linked stakeholder needs with materials properties and considered long-term system implications.

9. Creating Value (EM)

The team focused on improving usability while maintaining technical performance.

10. Feedback Addressed

Instructor feedback was incorporated into the next iteration cycle and influenced design decisions.

Report 70

1. Group Number and Title: Group 23 – Low-Cost Water Filtration Membrane

2. Date: February 12, 2026

3. HCD Spaces Engaged: Prototype, Synthesize

4. Activities Completed

- Activity 1: Conducted work related to Prototype space including data collection, analysis, concept refinement, or material testing.
- Activity 2: Conducted work related to Prototype space including data collection, analysis, concept refinement, or material testing.
- Activity 3: Conducted work related to Synthesize space including data collection, analysis, concept refinement, or material testing.
- Activity 4: Conducted work related to Prototype space including data collection, analysis, concept refinement, or material testing.
- Activity 5: Conducted work related to Synthesize space including data collection, analysis, concept refinement, or material testing.
- Activity 6: Conducted work related to Synthesize space including data collection, analysis, concept refinement, or material testing.

5. Expanded Description

The team deeply integrated stakeholder insights with materials testing. Iterative prototyping informed evidence-based decisions. Trade-offs between performance, manufacturability, and user needs were systematically evaluated.

6. Outstanding Challenges

- Address manufacturability risks.
- Improve prototype durability.
- Need additional testing data.

7. Curiosity Development (EM)

The team demonstrated inquisitiveness, contrarian thinking, experimentation, and comfort with ambiguity while iterating multiple prototype pathways.

8. Making Connections (EM)

The team applied systems thinking, creativity, and strategic risk management to integrate user insights, engineering constraints, and scalability considerations.

9. Creating Value (EM)

The team demonstrated customer-centric thinking, value awareness, persistence, and continuous improvement while refining prototypes to maximize user impact.

10. Feedback Addressed

Instructor feedback was incorporated into the next iteration cycle and influenced design decisions.

Report 71

1. Group Number and Title: Group 23 – Low-Cost Water Filtration Membrane

2. Date: March 16, 2026

3. HCD Spaces Engaged: Implement, Prototype

4. Activities Completed

- Activity 1: Conducted work related to Prototype space including data collection, analysis, concept refinement, or material testing.
- Activity 2: Conducted work related to Implement space including data collection, analysis, concept refinement, or material testing.
- Activity 3: Conducted work related to Implement space including data collection, analysis, concept refinement, or material testing.
- Activity 4: Conducted work related to Prototype space including data collection, analysis, concept refinement, or material testing.

5. Expanded Description

The team worked on assigned tasks and made incremental progress. Limited stakeholder engagement occurred. Decisions were primarily based on technical feasibility.

6. Outstanding Challenges

- Improve prototype durability.
- Address manufacturability risks.
- Increase stakeholder validation.

7. Curiosity Development (EM)

The team asked clarification questions and followed provided guidance.

8. Making Connections (EM)

Connections were primarily within course materials.

9. Creating Value (EM)

Progress contributed to completing project milestones.

10. Feedback Addressed

Instructor feedback was incorporated into the next iteration cycle and influenced design decisions.

Report 72

1. Group Number and Title: Group 10 – Smart Bio-Composite Packaging

2. Date: February 23, 2026

3. HCD Spaces Engaged: Synthesize, Prototype

4. Activities Completed

- Activity 1: Conducted work related to Prototype space including data collection, analysis, concept refinement, or material testing.
- Activity 2: Conducted work related to Prototype space including data collection, analysis, concept refinement, or material testing.
- Activity 3: Conducted work related to Prototype space including data collection, analysis, concept refinement, or material testing.
- Activity 4: Conducted work related to Prototype space including data collection, analysis, concept refinement, or material testing.

5. Expanded Description

The team worked on assigned tasks and made incremental progress. Limited stakeholder engagement occurred. Decisions were primarily based on technical feasibility.

6. Outstanding Challenges

- Improve prototype durability.
- Clarify cost constraints.
- Increase stakeholder validation.

7. Curiosity Development (EM)

The team asked clarification questions and followed provided guidance.

8. Making Connections (EM)

Connections were primarily within course materials.

9. Creating Value (EM)

Progress contributed to completing project milestones.

10. Feedback Addressed

Instructor feedback was incorporated into the next iteration cycle and influenced design decisions.

Report 73

1. Group Number and Title: Group 7 – Phase-Change Cooling Vest

2. Date: February 24, 2026

3. HCD Spaces Engaged: Understand, Synthesize, Prototype, Ideate, Implement

4. Activities Completed

- Activity 1: Conducted work related to Ideate space including data collection, analysis, concept refinement, or material testing.
- Activity 2: Conducted work related to Understand space including data collection, analysis, concept refinement, or material testing.
- Activity 3: Conducted work related to Understand space including data collection, analysis, concept refinement, or material testing.
- Activity 4: Conducted work related to Prototype space including data collection, analysis, concept refinement, or material testing.
- Activity 5: Conducted work related to Synthesize space including data collection, analysis, concept refinement, or material testing.
- Activity 6: Conducted work related to Prototype space including data collection, analysis, concept refinement, or material testing.

5. Expanded Description

The team balanced technical testing with user considerations. Some stakeholder feedback informed concept refinement. Iterative testing improved performance metrics.

6. Outstanding Challenges

- Address manufacturability risks.
- Clarify cost constraints.
- Increase stakeholder validation.
- Need additional testing data.

7. Curiosity Development (EM)

The team explored alternative material configurations and questioned initial assumptions about performance trade-offs.

8. Making Connections (EM)

The team linked stakeholder needs with materials properties and considered long-term system implications.

9. Creating Value (EM)

The team focused on improving usability while maintaining technical performance.

10. Feedback Addressed

Instructor feedback was incorporated into the next iteration cycle and influenced design decisions.

Report 74

1. Group Number and Title: Group 15 – Low-Cost Water Filtration Membrane

2. Date: February 20, 2026

3. HCD Spaces Engaged: Understand, Synthesize, Implement, Prototype

4. Activities Completed

- Activity 1: Conducted work related to Implement space including data collection, analysis, concept refinement, or material testing.
- Activity 2: Conducted work related to Synthesize space including data collection, analysis, concept refinement, or material testing.
- Activity 3: Conducted work related to Synthesize space including data collection, analysis, concept refinement, or material testing.

5. Expanded Description

The team deeply integrated stakeholder insights with materials testing. Iterative prototyping informed evidence-based decisions. Trade-offs between performance, manufacturability, and user needs were systematically evaluated.

6. Outstanding Challenges

- Clarify cost constraints.
- Need additional testing data.

7. Curiosity Development (EM)

The team demonstrated inquisitiveness, contrarian thinking, experimentation, and comfort with ambiguity while iterating multiple prototype pathways.

8. Making Connections (EM)

The team applied systems thinking, creativity, and strategic risk management to integrate user insights, engineering constraints, and scalability considerations.

9. Creating Value (EM)

The team demonstrated customer-centric thinking, value awareness, persistence, and continuous improvement while refining prototypes to maximize user impact.

10. Feedback Addressed

Instructor feedback was incorporated into the next iteration cycle and influenced design decisions.

Report 75

1. Group Number and Title: Group 16 – Lightweight Aerospace Alloy Redesign

2. Date: March 17, 2026

3. HCD Spaces Engaged: Understand, Implement, Synthesize, Prototype

4. Activities Completed

- Activity 1: Conducted work related to Implement space including data collection, analysis, concept refinement, or material testing.
- Activity 2: Conducted work related to Prototype space including data collection, analysis, concept refinement, or material testing.
- Activity 3: Conducted work related to Synthesize space including data collection, analysis, concept refinement, or material testing.
- Activity 4: Conducted work related to Understand space including data collection, analysis, concept refinement, or material testing.
- Activity 5: Conducted work related to Prototype space including data collection, analysis, concept refinement, or material testing.

5. Expanded Description

The team worked on assigned tasks and made incremental progress. Limited stakeholder engagement occurred. Decisions were primarily based on technical feasibility.

6. Outstanding Challenges

- Clarify cost constraints.
- Improve prototype durability.

7. Curiosity Development (EM)

The team asked clarification questions and followed provided guidance.

8. Making Connections (EM)

Connections were primarily within course materials.

9. Creating Value (EM)

Progress contributed to completing project milestones.

10. Feedback Addressed

Instructor feedback was incorporated into the next iteration cycle and influenced design decisions.

Report 76

1. Group Number and Title: Group 23 – Low-Cost Water Filtration Membrane

2. Date: March 02, 2026

3. HCD Spaces Engaged: Understand, Ideate, Synthesize, Prototype, Implement

4. Activities Completed

- Activity 1: Conducted work related to Implement space including data collection, analysis, concept refinement, or material testing.
- Activity 2: Conducted work related to Synthesize space including data collection, analysis, concept refinement, or material testing.
- Activity 3: Conducted work related to Understand space including data collection, analysis, concept refinement, or material testing.
- Activity 4: Conducted work related to Ideate space including data collection, analysis, concept refinement, or material testing.

5. Expanded Description

The team balanced technical testing with user considerations. Some stakeholder feedback informed concept refinement. Iterative testing improved performance metrics.

6. Outstanding Challenges

- Need additional testing data.
- Increase stakeholder validation.
- Address manufacturability risks.
- Improve prototype durability.

7. Curiosity Development (EM)

The team explored alternative material configurations and questioned initial assumptions about performance trade-offs.

8. Making Connections (EM)

The team linked stakeholder needs with materials properties and considered long-term system implications.

9. Creating Value (EM)

The team focused on improving usability while maintaining technical performance.

10. Feedback Addressed

Instructor feedback was incorporated into the next iteration cycle and influenced design decisions.

Report 77

1. Group Number and Title: Group 10 – Sustainable Concrete Additive

2. Date: February 16, 2026

3. HCD Spaces Engaged: Implement, Ideate, Prototype, Synthesize

4. Activities Completed

- Activity 1: Conducted work related to Prototype space including data collection, analysis, concept refinement, or material testing.
- Activity 2: Conducted work related to Implement space including data collection, analysis, concept refinement, or material testing.
- Activity 3: Conducted work related to Prototype space including data collection, analysis, concept refinement, or material testing.
- Activity 4: Conducted work related to Implement space including data collection, analysis, concept refinement, or material testing.
- Activity 5: Conducted work related to Synthesize space including data collection, analysis, concept refinement, or material testing.

5. Expanded Description

The team balanced technical testing with user considerations. Some stakeholder feedback informed concept refinement. Iterative testing improved performance metrics.

6. Outstanding Challenges

- Clarify cost constraints.
- Address manufacturability risks.
- Need additional testing data.

7. Curiosity Development (EM)

The team explored alternative material configurations and questioned initial assumptions about performance trade-offs.

8. Making Connections (EM)

The team linked stakeholder needs with materials properties and considered long-term system implications.

9. Creating Value (EM)

The team focused on improving usability while maintaining technical performance.

10. Feedback Addressed

Instructor feedback was incorporated into the next iteration cycle and influenced design decisions.

Report 78

1. Group Number and Title: Group 4 – Antimicrobial Hospital Surface Coating

2. Date: February 02, 2026

3. HCD Spaces Engaged: Implement, Understand, Ideate

4. Activities Completed

- Activity 1: Conducted work related to Ideate space including data collection, analysis, concept refinement, or material testing.
- Activity 2: Conducted work related to Implement space including data collection, analysis, concept refinement, or material testing.
- Activity 3: Conducted work related to Understand space including data collection, analysis, concept refinement, or material testing.
- Activity 4: Conducted work related to Understand space including data collection, analysis, concept refinement, or material testing.
- Activity 5: Conducted work related to Ideate space including data collection, analysis, concept refinement, or material testing.

5. Expanded Description

The team worked on assigned tasks and made incremental progress. Limited stakeholder engagement occurred. Decisions were primarily based on technical feasibility.

6. Outstanding Challenges

- Clarify cost constraints.
- Increase stakeholder validation.
- Need additional testing data.

7. Curiosity Development (EM)

The team asked clarification questions and followed provided guidance.

8. Making Connections (EM)

Connections were primarily within course materials.

9. Creating Value (EM)

Progress contributed to completing project milestones.

10. Feedback Addressed

Instructor feedback was incorporated into the next iteration cycle and influenced design decisions.

Report 79

1. Group Number and Title: Group 2 – Phase-Change Cooling Vest

2. Date: February 22, 2026

3. HCD Spaces Engaged: Prototype, Ideate, Synthesize, Implement, Understand

4. Activities Completed

- Activity 1: Conducted work related to Synthesize space including data collection, analysis, concept refinement, or material testing.
- Activity 2: Conducted work related to Synthesize space including data collection, analysis, concept refinement, or material testing.
- Activity 3: Conducted work related to Synthesize space including data collection, analysis, concept refinement, or material testing.
- Activity 4: Conducted work related to Ideate space including data collection, analysis, concept refinement, or material testing.

5. Expanded Description

The team balanced technical testing with user considerations. Some stakeholder feedback informed concept refinement. Iterative testing improved performance metrics.

6. Outstanding Challenges

- Address manufacturability risks.
- Need additional testing data.

7. Curiosity Development (EM)

The team explored alternative material configurations and questioned initial assumptions about performance trade-offs.

8. Making Connections (EM)

The team linked stakeholder needs with materials properties and considered long-term system implications.

9. Creating Value (EM)

The team focused on improving usability while maintaining technical performance.

10. Feedback Addressed

Instructor feedback was incorporated into the next iteration cycle and influenced design decisions.

Report 80

1. Group Number and Title: Group 24 – Smart Bio-Composite Packaging

2. Date: March 07, 2026

3. HCD Spaces Engaged: Ideate, Implement, Understand, Synthesize

4. Activities Completed

- Activity 1: Conducted work related to Implement space including data collection, analysis, concept refinement, or material testing.
- Activity 2: Conducted work related to Understand space including data collection, analysis, concept refinement, or material testing.
- Activity 3: Conducted work related to Ideate space including data collection, analysis, concept refinement, or material testing.

5. Expanded Description

The team balanced technical testing with user considerations. Some stakeholder feedback informed concept refinement. Iterative testing improved performance metrics.

6. Outstanding Challenges

- Address manufacturability risks.
- Improve prototype durability.
- Need additional testing data.
- Increase stakeholder validation.

7. Curiosity Development (EM)

The team explored alternative material configurations and questioned initial assumptions about performance trade-offs.

8. Making Connections (EM)

The team linked stakeholder needs with materials properties and considered long-term system implications.

9. Creating Value (EM)

The team focused on improving usability while maintaining technical performance.

10. Feedback Addressed

Instructor feedback was incorporated into the next iteration cycle and influenced design decisions.

Report 81

1. Group Number and Title: Group 1 – Smart Bio-Composite Packaging

2. Date: February 16, 2026

3. HCD Spaces Engaged: Prototype, Implement, Understand, Ideate

4. Activities Completed

- Activity 1: Conducted work related to Prototype space including data collection, analysis, concept refinement, or material testing.
- Activity 2: Conducted work related to Ideate space including data collection, analysis, concept refinement, or material testing.
- Activity 3: Conducted work related to Understand space including data collection, analysis, concept refinement, or material testing.
- Activity 4: Conducted work related to Ideate space including data collection, analysis, concept refinement, or material testing.
- Activity 5: Conducted work related to Prototype space including data collection, analysis, concept refinement, or material testing.
- Activity 6: Conducted work related to Ideate space including data collection, analysis, concept refinement, or material testing.

5. Expanded Description

The team balanced technical testing with user considerations. Some stakeholder feedback informed concept refinement. Iterative testing improved performance metrics.

6. Outstanding Challenges

- Improve prototype durability.
- Address manufacturability risks.
- Increase stakeholder validation.

7. Curiosity Development (EM)

The team explored alternative material configurations and questioned initial assumptions about performance trade-offs.

8. Making Connections (EM)

The team linked stakeholder needs with materials properties and considered long-term system implications.

9. Creating Value (EM)

The team focused on improving usability while maintaining technical performance.

10. Feedback Addressed

Instructor feedback was incorporated into the next iteration cycle and influenced design decisions.

Report 82

1. Group Number and Title: Group 21 – Lightweight Aerospace Alloy Redesign

2. Date: March 07, 2026

3. HCD Spaces Engaged: Understand, Ideate, Synthesize, Prototype

4. Activities Completed

- Activity 1: Conducted work related to Ideate space including data collection, analysis, concept refinement, or material testing.
- Activity 2: Conducted work related to Prototype space including data collection, analysis, concept refinement, or material testing.
- Activity 3: Conducted work related to Synthesize space including data collection, analysis, concept refinement, or material testing.
- Activity 4: Conducted work related to Ideate space including data collection, analysis, concept refinement, or material testing.
- Activity 5: Conducted work related to Understand space including data collection, analysis, concept refinement, or material testing.

5. Expanded Description

The team deeply integrated stakeholder insights with materials testing. Iterative prototyping informed evidence-based decisions. Trade-offs between performance, manufacturability, and user needs were systematically evaluated.

6. Outstanding Challenges

- Improve prototype durability.
- Need additional testing data.

7. Curiosity Development (EM)

The team demonstrated inquisitiveness, contrarian thinking, experimentation, and comfort with ambiguity while iterating multiple prototype pathways.

8. Making Connections (EM)

The team applied systems thinking, creativity, and strategic risk management to integrate user insights, engineering constraints, and scalability considerations.

9. Creating Value (EM)

The team demonstrated customer-centric thinking, value awareness, persistence, and continuous improvement while refining prototypes to maximize user impact.

10. Feedback Addressed

Instructor feedback was incorporated into the next iteration cycle and influenced design decisions.

Report 83

1. Group Number and Title: Group 6 – Antimicrobial Hospital Surface Coating

2. Date: March 07, 2026

3. HCD Spaces Engaged: Implement, Prototype, Ideate, Understand

4. Activities Completed

- Activity 1: Conducted work related to Ideate space including data collection, analysis, concept refinement, or material testing.
- Activity 2: Conducted work related to Understand space including data collection, analysis, concept refinement, or material testing.
- Activity 3: Conducted work related to Implement space including data collection, analysis, concept refinement, or material testing.
- Activity 4: Conducted work related to Prototype space including data collection, analysis, concept refinement, or material testing.
- Activity 5: Conducted work related to Ideate space including data collection, analysis, concept refinement, or material testing.

5. Expanded Description

The team worked on assigned tasks and made incremental progress. Limited stakeholder engagement occurred. Decisions were primarily based on technical feasibility.

6. Outstanding Challenges

- Improve prototype durability.
- Clarify cost constraints.
- Address manufacturability risks.

7. Curiosity Development (EM)

The team asked clarification questions and followed provided guidance.

8. Making Connections (EM)

Connections were primarily within course materials.

9. Creating Value (EM)

Progress contributed to completing project milestones.

10. Feedback Addressed

Instructor feedback was incorporated into the next iteration cycle and influenced design decisions.

Report 84

1. Group Number and Title: Group 8 – Smart Bio-Composite Packaging

2. Date: February 08, 2026

3. HCD Spaces Engaged: Understand, Ideate, Synthesize, Implement

4. Activities Completed

- Activity 1: Conducted work related to Understand space including data collection, analysis, concept refinement, or material testing.
- Activity 2: Conducted work related to Implement space including data collection, analysis, concept refinement, or material testing.
- Activity 3: Conducted work related to Ideate space including data collection, analysis, concept refinement, or material testing.
- Activity 4: Conducted work related to Implement space including data collection, analysis, concept refinement, or material testing.
- Activity 5: Conducted work related to Understand space including data collection, analysis, concept refinement, or material testing.

5. Expanded Description

The team worked on assigned tasks and made incremental progress. Limited stakeholder engagement occurred. Decisions were primarily based on technical feasibility.

6. Outstanding Challenges

- Increase stakeholder validation.
- Improve prototype durability.
- Need additional testing data.

7. Curiosity Development (EM)

The team asked clarification questions and followed provided guidance.

8. Making Connections (EM)

Connections were primarily within course materials.

9. Creating Value (EM)

Progress contributed to completing project milestones.

10. Feedback Addressed

Instructor feedback was incorporated into the next iteration cycle and influenced design decisions.

Report 85

1. Group Number and Title: Group 9 – Smart Bio-Composite Packaging

2. Date: March 26, 2026

3. HCD Spaces Engaged: Implement, Understand, Ideate, Synthesize

4. Activities Completed

- Activity 1: Conducted work related to Synthesize space including data collection, analysis, concept refinement, or material testing.
- Activity 2: Conducted work related to Ideate space including data collection, analysis, concept refinement, or material testing.
- Activity 3: Conducted work related to Ideate space including data collection, analysis, concept refinement, or material testing.
- Activity 4: Conducted work related to Ideate space including data collection, analysis, concept refinement, or material testing.
- Activity 5: Conducted work related to Understand space including data collection, analysis, concept refinement, or material testing.
- Activity 6: Conducted work related to Ideate space including data collection, analysis, concept refinement, or material testing.

5. Expanded Description

The team balanced technical testing with user considerations. Some stakeholder feedback informed concept refinement. Iterative testing improved performance metrics.

6. Outstanding Challenges

- Need additional testing data.
- Address manufacturability risks.
- Improve prototype durability.

7. Curiosity Development (EM)

The team explored alternative material configurations and questioned initial assumptions about performance trade-offs.

8. Making Connections (EM)

The team linked stakeholder needs with materials properties and considered long-term system implications.

9. Creating Value (EM)

The team focused on improving usability while maintaining technical performance.

10. Feedback Addressed

Instructor feedback was incorporated into the next iteration cycle and influenced design decisions.

Report 86

1. Group Number and Title: Group 14 – Low-Cost Water Filtration Membrane

2. Date: March 18, 2026

3. HCD Spaces Engaged: Understand, Ideate, Synthesize, Prototype, Implement

4. Activities Completed

- Activity 1: Conducted work related to Synthesize space including data collection, analysis, concept refinement, or material testing.
- Activity 2: Conducted work related to Ideate space including data collection, analysis, concept refinement, or material testing.
- Activity 3: Conducted work related to Implement space including data collection, analysis, concept refinement, or material testing.
- Activity 4: Conducted work related to Ideate space including data collection, analysis, concept refinement, or material testing.
- Activity 5: Conducted work related to Understand space including data collection, analysis, concept refinement, or material testing.

5. Expanded Description

The team deeply integrated stakeholder insights with materials testing. Iterative prototyping informed evidence-based decisions. Trade-offs between performance, manufacturability, and user needs were systematically evaluated.

6. Outstanding Challenges

- Need additional testing data.
- Clarify cost constraints.
- Improve prototype durability.

7. Curiosity Development (EM)

The team demonstrated inquisitiveness, contrarian thinking, experimentation, and comfort with ambiguity while iterating multiple prototype pathways.

8. Making Connections (EM)

The team applied systems thinking, creativity, and strategic risk management to integrate user insights, engineering constraints, and scalability considerations.

9. Creating Value (EM)

The team demonstrated customer-centric thinking, value awareness, persistence, and continuous improvement while refining prototypes to maximize user impact.

10. Feedback Addressed

Instructor feedback was incorporated into the next iteration cycle and influenced design decisions.

Report 87

1. Group Number and Title: Group 14 – Phase-Change Cooling Vest

2. Date: March 04, 2026

3. HCD Spaces Engaged: Synthesize, Prototype

4. Activities Completed

- Activity 1: Conducted work related to Synthesize space including data collection, analysis, concept refinement, or material testing.
- Activity 2: Conducted work related to Prototype space including data collection, analysis, concept refinement, or material testing.
- Activity 3: Conducted work related to Synthesize space including data collection, analysis, concept refinement, or material testing.
- Activity 4: Conducted work related to Prototype space including data collection, analysis, concept refinement, or material testing.
- Activity 5: Conducted work related to Prototype space including data collection, analysis, concept refinement, or material testing.
- Activity 6: Conducted work related to Prototype space including data collection, analysis, concept refinement, or material testing.

5. Expanded Description

The team worked on assigned tasks and made incremental progress. Limited stakeholder engagement occurred. Decisions were primarily based on technical feasibility.

6. Outstanding Challenges

- Address manufacturability risks.
- Clarify cost constraints.
- Improve prototype durability.

7. Curiosity Development (EM)

The team asked clarification questions and followed provided guidance.

8. Making Connections (EM)

Connections were primarily within course materials.

9. Creating Value (EM)

Progress contributed to completing project milestones.

10. Feedback Addressed

Instructor feedback was incorporated into the next iteration cycle and influenced design decisions.

Report 88

1. Group Number and Title: Group 6 – Low-Cost Water Filtration Membrane

2. Date: February 14, 2026

3. HCD Spaces Engaged: Prototype, Synthesize, Ideate, Implement, Understand

4. Activities Completed

- Activity 1: Conducted work related to Implement space including data collection, analysis, concept refinement, or material testing.
- Activity 2: Conducted work related to Synthesize space including data collection, analysis, concept refinement, or material testing.
- Activity 3: Conducted work related to Prototype space including data collection, analysis, concept refinement, or material testing.
- Activity 4: Conducted work related to Prototype space including data collection, analysis, concept refinement, or material testing.
- Activity 5: Conducted work related to Synthesize space including data collection, analysis, concept refinement, or material testing.
- Activity 6: Conducted work related to Implement space including data collection, analysis, concept refinement, or material testing.

5. Expanded Description

The team deeply integrated stakeholder insights with materials testing. Iterative prototyping informed evidence-based decisions. Trade-offs between performance, manufacturability, and user needs were systematically evaluated.

6. Outstanding Challenges

- Address manufacturability risks.
- Increase stakeholder validation.
- Improve prototype durability.

7. Curiosity Development (EM)

The team demonstrated inquisitiveness, contrarian thinking, experimentation, and comfort with ambiguity while iterating multiple prototype pathways.

8. Making Connections (EM)

The team applied systems thinking, creativity, and strategic risk management to integrate user insights, engineering constraints, and scalability considerations.

9. Creating Value (EM)

The team demonstrated customer-centric thinking, value awareness, persistence, and continuous improvement while refining prototypes to maximize user impact.

10. Feedback Addressed

Instructor feedback was incorporated into the next iteration cycle and influenced design decisions.

Report 89

1. Group Number and Title: Group 24 – Smart Bio-Composite Packaging

2. Date: March 17, 2026

3. HCD Spaces Engaged: Synthesize, Implement, Understand, Ideate, Prototype

4. Activities Completed

- Activity 1: Conducted work related to Prototype space including data collection, analysis, concept refinement, or material testing.
- Activity 2: Conducted work related to Synthesize space including data collection, analysis, concept refinement, or material testing.
- Activity 3: Conducted work related to Understand space including data collection, analysis, concept refinement, or material testing.

5. Expanded Description

The team worked on assigned tasks and made incremental progress. Limited stakeholder engagement occurred. Decisions were primarily based on technical feasibility.

6. Outstanding Challenges

- Address manufacturability risks.
- Improve prototype durability.
- Increase stakeholder validation.
- Need additional testing data.

7. Curiosity Development (EM)

The team asked clarification questions and followed provided guidance.

8. Making Connections (EM)

Connections were primarily within course materials.

9. Creating Value (EM)

Progress contributed to completing project milestones.

10. Feedback Addressed

Instructor feedback was incorporated into the next iteration cycle and influenced design decisions.

Report 90

1. Group Number and Title: Group 15 – Sustainable Concrete Additive

2. Date: March 10, 2026

3. HCD Spaces Engaged: Ideate, Understand, Implement, Synthesize, Prototype

4. Activities Completed

- Activity 1: Conducted work related to Synthesize space including data collection, analysis, concept refinement, or material testing.
- Activity 2: Conducted work related to Ideate space including data collection, analysis, concept refinement, or material testing.
- Activity 3: Conducted work related to Synthesize space including data collection, analysis, concept refinement, or material testing.
- Activity 4: Conducted work related to Understand space including data collection, analysis, concept refinement, or material testing.

5. Expanded Description

The team deeply integrated stakeholder insights with materials testing. Iterative prototyping informed evidence-based decisions. Trade-offs between performance, manufacturability, and user needs were systematically evaluated.

6. Outstanding Challenges

- Address manufacturability risks.
- Improve prototype durability.
- Clarify cost constraints.

7. Curiosity Development (EM)

The team demonstrated inquisitiveness, contrarian thinking, experimentation, and comfort with ambiguity while iterating multiple prototype pathways.

8. Making Connections (EM)

The team applied systems thinking, creativity, and strategic risk management to integrate user insights, engineering constraints, and scalability considerations.

9. Creating Value (EM)

The team demonstrated customer-centric thinking, value awareness, persistence, and continuous improvement while refining prototypes to maximize user impact.

10. Feedback Addressed

Instructor feedback was incorporated into the next iteration cycle and influenced design decisions.

Report 91

1. Group Number and Title: Group 25 – Recyclable Polymer Blend Development

2. Date: February 07, 2026

3. HCD Spaces Engaged: Prototype, Implement, Synthesize, Understand, Ideate

4. Activities Completed

- Activity 1: Conducted work related to Implement space including data collection, analysis, concept refinement, or material testing.
- Activity 2: Conducted work related to Ideate space including data collection, analysis, concept refinement, or material testing.
- Activity 3: Conducted work related to Prototype space including data collection, analysis, concept refinement, or material testing.
- Activity 4: Conducted work related to Ideate space including data collection, analysis, concept refinement, or material testing.
- Activity 5: Conducted work related to Synthesize space including data collection, analysis, concept refinement, or material testing.

5. Expanded Description

The team worked on assigned tasks and made incremental progress. Limited stakeholder engagement occurred. Decisions were primarily based on technical feasibility.

6. Outstanding Challenges

- Address manufacturability risks.
- Clarify cost constraints.
- Need additional testing data.
- Increase stakeholder validation.

7. Curiosity Development (EM)

The team asked clarification questions and followed provided guidance.

8. Making Connections (EM)

Connections were primarily within course materials.

9. Creating Value (EM)

Progress contributed to completing project milestones.

10. Feedback Addressed

Instructor feedback was incorporated into the next iteration cycle and influenced design decisions.

Report 92

1. Group Number and Title: Group 1 – Sustainable Concrete Additive

2. Date: March 31, 2026

3. HCD Spaces Engaged: Prototype, Synthesize, Implement, Ideate

4. Activities Completed

- Activity 1: Conducted work related to Synthesize space including data collection, analysis, concept refinement, or material testing.
- Activity 2: Conducted work related to Ideate space including data collection, analysis, concept refinement, or material testing.
- Activity 3: Conducted work related to Ideate space including data collection, analysis, concept refinement, or material testing.
- Activity 4: Conducted work related to Synthesize space including data collection, analysis, concept refinement, or material testing.
- Activity 5: Conducted work related to Synthesize space including data collection, analysis, concept refinement, or material testing.

5. Expanded Description

The team worked on assigned tasks and made incremental progress. Limited stakeholder engagement occurred. Decisions were primarily based on technical feasibility.

6. Outstanding Challenges

- Increase stakeholder validation.
- Improve prototype durability.
- Clarify cost constraints.
- Need additional testing data.

7. Curiosity Development (EM)

The team asked clarification questions and followed provided guidance.

8. Making Connections (EM)

Connections were primarily within course materials.

9. Creating Value (EM)

Progress contributed to completing project milestones.

10. Feedback Addressed

Instructor feedback was incorporated into the next iteration cycle and influenced design decisions.

Report 93

1. Group Number and Title: Group 5 – Antimicrobial Hospital Surface Coating

2. Date: April 01, 2026

3. HCD Spaces Engaged: Understand, Prototype, Implement, Synthesize

4. Activities Completed

- Activity 1: Conducted work related to Implement space including data collection, analysis, concept refinement, or material testing.
- Activity 2: Conducted work related to Synthesize space including data collection, analysis, concept refinement, or material testing.
- Activity 3: Conducted work related to Synthesize space including data collection, analysis, concept refinement, or material testing.
- Activity 4: Conducted work related to Understand space including data collection, analysis, concept refinement, or material testing.
- Activity 5: Conducted work related to Synthesize space including data collection, analysis, concept refinement, or material testing.

5. Expanded Description

The team balanced technical testing with user considerations. Some stakeholder feedback informed concept refinement. Iterative testing improved performance metrics.

6. Outstanding Challenges

- Clarify cost constraints.
- Address manufacturability risks.
- Improve prototype durability.
- Increase stakeholder validation.

7. Curiosity Development (EM)

The team explored alternative material configurations and questioned initial assumptions about performance trade-offs.

8. Making Connections (EM)

The team linked stakeholder needs with materials properties and considered long-term system implications.

9. Creating Value (EM)

The team focused on improving usability while maintaining technical performance.

10. Feedback Addressed

Instructor feedback was incorporated into the next iteration cycle and influenced design decisions.

Report 94

1. Group Number and Title: Group 24 – Wearable Thermal Regulation System

2. Date: February 22, 2026

3. HCD Spaces Engaged: Understand, Implement, Synthesize, Prototype, Ideate

4. Activities Completed

- Activity 1: Conducted work related to Implement space including data collection, analysis, concept refinement, or material testing.
- Activity 2: Conducted work related to Prototype space including data collection, analysis, concept refinement, or material testing.
- Activity 3: Conducted work related to Understand space including data collection, analysis, concept refinement, or material testing.

5. Expanded Description

The team balanced technical testing with user considerations. Some stakeholder feedback informed concept refinement. Iterative testing improved performance metrics.

6. Outstanding Challenges

- Clarify cost constraints.
- Address manufacturability risks.
- Need additional testing data.

7. Curiosity Development (EM)

The team explored alternative material configurations and questioned initial assumptions about performance trade-offs.

8. Making Connections (EM)

The team linked stakeholder needs with materials properties and considered long-term system implications.

9. Creating Value (EM)

The team focused on improving usability while maintaining technical performance.

10. Feedback Addressed

Instructor feedback was incorporated into the next iteration cycle and influenced design decisions.

Report 95

1. Group Number and Title: Group 24 – Sustainable Concrete Additive

2. Date: March 31, 2026

3. HCD Spaces Engaged: Ideate, Prototype, Implement, Synthesize

4. Activities Completed

- Activity 1: Conducted work related to Implement space including data collection, analysis, concept refinement, or material testing.
- Activity 2: Conducted work related to Ideate space including data collection, analysis, concept refinement, or material testing.
- Activity 3: Conducted work related to Prototype space including data collection, analysis, concept refinement, or material testing.

5. Expanded Description

The team deeply integrated stakeholder insights with materials testing. Iterative prototyping informed evidence-based decisions. Trade-offs between performance, manufacturability, and user needs were systematically evaluated.

6. Outstanding Challenges

- Need additional testing data.
- Clarify cost constraints.
- Improve prototype durability.
- Address manufacturability risks.

7. Curiosity Development (EM)

The team demonstrated inquisitiveness, contrarian thinking, experimentation, and comfort with ambiguity while iterating multiple prototype pathways.

8. Making Connections (EM)

The team applied systems thinking, creativity, and strategic risk management to integrate user insights, engineering constraints, and scalability considerations.

9. Creating Value (EM)

The team demonstrated customer-centric thinking, value awareness, persistence, and continuous improvement while refining prototypes to maximize user impact.

10. Feedback Addressed

Instructor feedback was incorporated into the next iteration cycle and influenced design decisions.

Report 96

1. Group Number and Title: Group 8 – Wearable Thermal Regulation System

2. Date: March 15, 2026

3. HCD Spaces Engaged: Understand, Prototype, Synthesize, Implement

4. Activities Completed

- Activity 1: Conducted work related to Implement space including data collection, analysis, concept refinement, or material testing.
- Activity 2: Conducted work related to Synthesize space including data collection, analysis, concept refinement, or material testing.
- Activity 3: Conducted work related to Synthesize space including data collection, analysis, concept refinement, or material testing.
- Activity 4: Conducted work related to Synthesize space including data collection, analysis, concept refinement, or material testing.

5. Expanded Description

The team balanced technical testing with user considerations. Some stakeholder feedback informed concept refinement. Iterative testing improved performance metrics.

6. Outstanding Challenges

- Increase stakeholder validation.
- Clarify cost constraints.
- Address manufacturability risks.

7. Curiosity Development (EM)

The team explored alternative material configurations and questioned initial assumptions about performance trade-offs.

8. Making Connections (EM)

The team linked stakeholder needs with materials properties and considered long-term system implications.

9. Creating Value (EM)

The team focused on improving usability while maintaining technical performance.

10. Feedback Addressed

Instructor feedback was incorporated into the next iteration cycle and influenced design decisions.

Report 97

1. Group Number and Title: Group 25 – Sustainable Concrete Additive

2. Date: March 20, 2026

3. HCD Spaces Engaged: Understand, Ideate, Prototype, Implement, Synthesize

4. Activities Completed

- Activity 1: Conducted work related to Prototype space including data collection, analysis, concept refinement, or material testing.
- Activity 2: Conducted work related to Understand space including data collection, analysis, concept refinement, or material testing.
- Activity 3: Conducted work related to Understand space including data collection, analysis, concept refinement, or material testing.
- Activity 4: Conducted work related to Synthesize space including data collection, analysis, concept refinement, or material testing.

5. Expanded Description

The team worked on assigned tasks and made incremental progress. Limited stakeholder engagement occurred. Decisions were primarily based on technical feasibility.

6. Outstanding Challenges

- Clarify cost constraints.
- Increase stakeholder validation.
- Address manufacturability risks.
- Need additional testing data.

7. Curiosity Development (EM)

The team asked clarification questions and followed provided guidance.

8. Making Connections (EM)

Connections were primarily within course materials.

9. Creating Value (EM)

Progress contributed to completing project milestones.

10. Feedback Addressed

Instructor feedback was incorporated into the next iteration cycle and influenced design decisions.

Report 98

1. Group Number and Title: Group 10 – Phase-Change Cooling Vest

2. Date: March 20, 2026

3. HCD Spaces Engaged: Implement, Synthesize, Understand

4. Activities Completed

- Activity 1: Conducted work related to Understand space including data collection, analysis, concept refinement, or material testing.
- Activity 2: Conducted work related to Synthesize space including data collection, analysis, concept refinement, or material testing.
- Activity 3: Conducted work related to Implement space including data collection, analysis, concept refinement, or material testing.
- Activity 4: Conducted work related to Synthesize space including data collection, analysis, concept refinement, or material testing.

5. Expanded Description

The team balanced technical testing with user considerations. Some stakeholder feedback informed concept refinement. Iterative testing improved performance metrics.

6. Outstanding Challenges

- Need additional testing data.
- Address manufacturability risks.

7. Curiosity Development (EM)

The team explored alternative material configurations and questioned initial assumptions about performance trade-offs.

8. Making Connections (EM)

The team linked stakeholder needs with materials properties and considered long-term system implications.

9. Creating Value (EM)

The team focused on improving usability while maintaining technical performance.

10. Feedback Addressed

Instructor feedback was incorporated into the next iteration cycle and influenced design decisions.

Report 99

1. Group Number and Title: Group 21 – Recyclable Polymer Blend Development

2. Date: February 24, 2026

3. HCD Spaces Engaged: Implement, Prototype

4. Activities Completed

- Activity 1: Conducted work related to Prototype space including data collection, analysis, concept refinement, or material testing.
- Activity 2: Conducted work related to Prototype space including data collection, analysis, concept refinement, or material testing.
- Activity 3: Conducted work related to Implement space including data collection, analysis, concept refinement, or material testing.
- Activity 4: Conducted work related to Implement space including data collection, analysis, concept refinement, or material testing.
- Activity 5: Conducted work related to Prototype space including data collection, analysis, concept refinement, or material testing.
- Activity 6: Conducted work related to Implement space including data collection, analysis, concept refinement, or material testing.

5. Expanded Description

The team balanced technical testing with user considerations. Some stakeholder feedback informed concept refinement. Iterative testing improved performance metrics.

6. Outstanding Challenges

- Need additional testing data.
- Clarify cost constraints.
- Address manufacturability risks.
- Improve prototype durability.

7. Curiosity Development (EM)

The team explored alternative material configurations and questioned initial assumptions about performance trade-offs.

8. Making Connections (EM)

The team linked stakeholder needs with materials properties and considered long-term system implications.

9. Creating Value (EM)

The team focused on improving usability while maintaining technical performance.

10. Feedback Addressed

Instructor feedback was incorporated into the next iteration cycle and influenced design decisions.

Report 100

1. Group Number and Title: Group 18 – Smart Bio-Composite Packaging

2. Date: March 29, 2026

3. HCD Spaces Engaged: Prototype, Ideate

4. Activities Completed

- Activity 1: Conducted work related to Ideate space including data collection, analysis, concept refinement, or material testing.
- Activity 2: Conducted work related to Prototype space including data collection, analysis, concept refinement, or material testing.
- Activity 3: Conducted work related to Prototype space including data collection, analysis, concept refinement, or material testing.
- Activity 4: Conducted work related to Prototype space including data collection, analysis, concept refinement, or material testing.
- Activity 5: Conducted work related to Prototype space including data collection, analysis, concept refinement, or material testing.
- Activity 6: Conducted work related to Prototype space including data collection, analysis, concept refinement, or material testing.

5. Expanded Description

The team balanced technical testing with user considerations. Some stakeholder feedback informed concept refinement. Iterative testing improved performance metrics.

6. Outstanding Challenges

- Increase stakeholder validation.
- Address manufacturability risks.

7. Curiosity Development (EM)

The team explored alternative material configurations and questioned initial assumptions about performance trade-offs.

8. Making Connections (EM)

The team linked stakeholder needs with materials properties and considered long-term system implications.

9. Creating Value (EM)

The team focused on improving usability while maintaining technical performance.

10. Feedback Addressed

Instructor feedback was incorporated into the next iteration cycle and influenced design decisions.